

Build Two-Way Phase Synchronization

Code it from the ground up — the snippets on these slides ARE the implementation

Distributed-array phase synchronization project

How this guide works

- You are going to write **one Python file**, `build_twoway.py`, from top to bottom. Each section of this deck explains what comes next, then gives you the code to type — **the snippets, typed in order, are the complete file**. Nothing is elided, nothing is toy code.
- And the file is not a simplification either: every line is **verbatim from the repository** (`ota_sync/core.py`, `sdr.py`, `coherent.py`); each section cites its source lines. When you finish, you have built the repository's actual two-way synchronization — and running your file reproduces the repository's output **bit for bit** (verified; the check is on the last slide).
- Requirements: Python with `torch` and `sionna` ≥ 2.0 .
- What you are building: two stations, each with its own drifting clock, exchange pilot waveforms both ways through a 3GPP multipath channel every 50 ms; a receiver chain measures timing/frequency/phase from raw samples; an extended Kalman filter tracks the clock offset; closed-loop corrections hold the pair aligned. Final performance, which your file will print: coherent gain 99.8%.

The build order

- 1 **Header** — imports and the two working dtypes.
- 2 **Small tools** — `wrap_phase`, covariance root, device pick.
- 3 **The Oscillator** — the drifting clock (the enemy).
- 4 **The config** — every physical assumption in one dataclass.
- 5 **The pilot** — Zadoff–Chu training fields.
- 6 **Hardware imperfections** — converters, amplifier, sample-clock, flicker noise.
- 7 **The radio link** — Sionna channel + the full capture chain.
- 8 **The receiver** — timing, frequency, phase from raw samples; R from physics.
- 9 **The filter** — the extended Kalman filter.
- 10 **The loop** — half-difference, branch handling, corrections, metrics.
- 11 **Run it** — and verify against the repository.

The file header

Create `build_twoway.py`. First the imports: `torch`, and four Sionna pieces — the noise block, the channel operator, the 3GPP tapped-delay-line model, and two helpers that convert a physical channel response into discrete-time taps. All numerics run in double precision: phase accumulates over millions of samples, and `float32` rounding would show up as fake oscillator noise.

The file header — type this

```
"""Two-way over-the-air phase synchronization, built from the ground up.
```

```
Every line below is verbatim from ota_sync/ (core.py, sdr.py, coherent.py) -- only the import header differs, because everything lives in this one file. Typing this file section by section, as the coding guide walks it, produces the repository's actual two-way implementation; running it reproduces the repository's numbers exactly.
```

```
"""
```

```
from __future__ import annotations
```

```
from dataclasses import dataclass
```

```
import math
```

```
from typing import Optional
```

```
import torch
```

```
from sionna.phy import config as sionna_config
```

```
from sionna.phy.channel import AWGN, ApplyTimeChannel
```

```
from sionna.phy.channel.tr38901 import TDL
```

```
from sionna.phy.channel.utils import (
```

```
    cir_to_time_channel,
```

```
    time_lag_discrete_time_channel,
```

```
)
```

```
REAL_DTYPE = torch.float64
```

```
COMPLEX_DTYPE = torch.complex128
```

A phase is an angle: 370° is 10° . Every phase subtraction in this program goes through this one-liner, or differences near the boundary come out wrong by a full turn. It appears on nearly every line that touches a phase; when debugging this kind of code, suspect a missing wrap first. (Repository: `core.py:24.`)

wrap_phase — type this

```
def wrap_phase(value: torch.Tensor) -> torch.Tensor:
    """Wrap angles in radians to  $[-\pi, \pi)$ ."""
    return torch remainder(value + math.pi, 2.0 * math.pi) - math.pi
```

Two small linear-algebra helpers

`_as_matrix` is convenience. `_covariance_root` matters: to draw correlated noise with covariance Q you need a matrix square root of Q ; an eigendecomposition gives one that tolerates positive-*semidefinite* Q (a zero eigenvalue = a noiseless direction), which a plain Cholesky factorization would reject. (`core.py:50--62.`)

Two small linear-algebra helpers — type this

```
def _as_matrix(
    value: torch.Tensor | list[list[float]], device: torch.device
) -> torch.Tensor:
    return torch.as_tensor(value, dtype=REAL_DTYPE, device=device)

def _covariance_root(covariance: torch.Tensor) -> torch.Tensor:
    """Return a square root for a positive-semidefinite covariance matrix."""
    eigenvalues, eigenvectors = torch.linalg.eigh(covariance)
    if torch.any(eigenvalues < -1e-12):
        raise ValueError("covariance must be positive semidefinite")
    return eigenvectors @ torch.diag(torch.sqrt(torch.clamp(eigenvalues, min=0.0)))
```

resolve_device

Runs on the graphics card when one is available, the processor otherwise.
(core.py:133.)

resolve_device — type this

```
def resolve_device(requested: str) -> torch.device:
    """Resolve ``auto`` to CUDA when available and CPU otherwise."""

    if requested == "auto":
        return torch.device("cuda" if torch.cuda.is_available() else "cpu")
    device = torch.device(requested)
    if device.type == "cuda" and not torch.cuda.is_available():
        raise RuntimeError("CUDA was requested but is not available")
    return device
```

The Oscillator – the enemy

The clock model: state $[\theta, \omega]$ (phase, frequency). The transition matrix says phase advances by frequency $\times$ time. The noise is a **random walk in both components** – drawn through the covariance root so the two components are correctly correlated – and random walks accumulate; they never average away. `apply_correction` is the actuator the loop will drive; nothing else may touch the state. Keep the transition matrix in mind: the Kalman filter carries an exact copy of it, so filter and physics agree by construction. (`core.py:175--225.`)

The Oscillator – the enemy — type this (1/2)

```
class Oscillator:
    """Random-walk phase and angular-frequency oscillator model."""

    def __init__(
        self,
        initial_phase: float,
        initial_frequency: float,
        sync_interval: float,
        process_covariance: torch.Tensor,
        device: torch.device,
        generator: Optional[torch.Generator] = None,
    ) -> None:
        self.device = device
        self.state = torch.tensor(
            [initial_phase, initial_frequency], dtype=REAL_DTYPE, device=device
        )
        self.transition = torch.tensor(
            [[1.0, sync_interval], [0.0, 1.0]],
            dtype=REAL_DTYPE,
            device=device,
        )
        self.process_covariance = _as_matrix(process_covariance, device)
        if self.process_covariance.shape != (2, 2):
            raise ValueError("process_covariance must have shape [2, 2]")
        self._noise_root = _covariance_root(self.process_covariance)

        self._generator = generator
```

The Oscillator – the enemy — type this (2/2)

```
@property
def phase(self) -> torch.Tensor:
    return self.state[0]

@property
def frequency(self) -> torch.Tensor:
    return self.state[1]

def step(self) -> torch.Tensor:
    white_noise = torch.randn(
        2,
        dtype=REAL_DTYPE,
        device=self.device,
        generator=self._generator,
    )
    process_noise = self._noise_root @ white_noise
    self.state = self.transition @ self.state + process_noise
    self.state[0] = wrap_phase(self.state[0])
    return self.state.clone()

def apply_correction(self, correction: torch.Tensor) -> None:
    self.state = self.state + correction.to(dtype=REAL_DTYPE, device=self.device)
    self.state[0] = wrap_phase(self.state[0])
```

SDRSimulationConfig – every physical assumption

One frozen dataclass holds every knob: timing (50 ms intervals, 1 MHz sampling), the channel (3GPP model “D”, 100 ns delay spread, 915 MHz), the pilot structure (16×16 short, 2047×2 long), the oscillator noise levels, every hardware impairment, and the loop’s actuation limits (16-bit phase corrections, one-interval command latency, 1 ms turnaround). Every experiment in the repository is a variation of these fields – never edited code. (`sdr.py:33--94.`)

SDRSimulationConfig – every physical assumption — type this (1/2)

```
@dataclass(frozen=True)
class SDRSimulationConfig:
    """Configuration of the sampled-IQ SDR synchronization experiment."""

    sync_interval: float = 0.05
    sample_rate: float = 1e6
    num_iterations: int = 100
    snr_db: float = 20.0
    carrier_frequency_hz: float = 915e6

    tdl_model: str = "D"
    delay_spread_s: float = 100e-9
    maximum_channel_delay_s: float = 3e-6
    channel_speed_mps: float = 0.0

    short_sequence_length: int = 16
    short_repetitions: int = 16
    long_sequence_length: int = 2047
    long_cp_length: int = 128
    long_repetitions: int = 2
    capture_guard_samples: int = 64
    timing_jitter_samples: int = 32

    master_initial_phase: float = 0.0
    master_initial_frequency_hz: float = 0.0
    slave_initial_phase: float = 1.2
    slave_initial_frequency_hz: float = 1500.0
    phase_process_std_rad: float = 0.002
    frequency_process_std_hz: float = 0.1
```


SDRSimulationConfig – every physical assumption — type this (2/2)

```
# None derives the sample-clock offset from the carrier offset (shared
# reference crystal); a number forces a fixed, independent offset.
sample_clock_offset_ppm: float | None = None
phase_noise_std_rad: float = 2e-4
phase_noise_white_pm_std_rad: float = 0.005
flicker_frequency_std_hz: float = 0.05
shadowing_std_db: float = 2.0
shadowing_correlation_s: float = 10.0
iq_gain_imbalance_db: float = 0.2
iq_phase_imbalance_deg: float = 1.0
dc_offset: complex = 0.005 + 0.003j
adc_bits: int = 12
dac_bits: int = 12
tx_amplitude: float = 0.5
pa_clip_level: float = 0.9
agc_target_rms: float = 0.25
detection_threshold: float = 0.25

phase_correction_bits: int = 16
frequency_correction_resolution_hz: float = 0.01
correction_latency_intervals: int = 1
# Residual TX/RX chain phase asymmetry between the nodes after loopback
# calibration; it does not cancel in a two-way exchange.
twoway_chain_asymmetry_deg: float = 0.0
# Half-duplex (TDD) turnaround inserted between the forward and reverse
# frames of a two-way exchange. The oscillators keep drifting across the
# gap; the loop compensates the deterministic part with its own CFO
# estimate, so the residual penalty is the frequency-estimate error
# times the gap plus the phase random walk accumulated during it.
tdd_turnaround_s: float = 1e-3
seed: int = 0
device: str = "auto"
```

Config validation

The dataclass rejects nonsense at construction: negative noise levels, a cyclic prefix longer than its sequence, too few training repeats. Fail loudly at setup, not silently mid-run. The `sample_period` property is the only derived quantity. (`sdr.py:96--157.`)

Config validation — type this (1/2)

```
def __post_init__(self) -> None:
    if self.sync_interval <= 0.0 or self.sample_rate <= 0.0:
        raise ValueError("sync_interval and sample_rate must be positive")
    if self.num_iterations < 1:
        raise ValueError("num_iterations must be positive")
    if self.carrier_frequency_hz <= 0.0:
        raise ValueError("carrier_frequency_hz must be positive")
    if self.tdl_model not in {
        "A",
        "B",
        "C",
        "D",
        "E",
        "A30",
        "B100",
        "C300",
    }:
        raise ValueError("invalid Sionna TDL model")
    if self.delay_spread_s <= 0.0 or self.maximum_channel_delay_s <= 0.0:
        raise ValueError("channel delays must be positive")
    if self.channel_speed_mps < 0.0:
        raise ValueError("channel_speed_mps cannot be negative")
    if self.short_sequence_length < 2 or self.short_repetitions < 3:
        raise ValueError("the short training field needs at least three repeats")
    if self.long_sequence_length < 3 or self.long_repetitions < 2:
        raise ValueError("the long training field needs at least two repeats")
    if not 0 <= self.long_cp_length < self.long_sequence_length:
        raise ValueError("long_cp_length must be shorter than the long sequence")
    if self.capture_guard_samples < 1 or self.timing_jitter_samples < 0:
        raise ValueError("capture guard must be positive and jitter nonnegative")
```

Config validation — type this (2/2)

```
if self.phase_process_std_rad < 0.0 or self.frequency_process_std_hz < 0.0:
    raise ValueError("oscillator process deviations cannot be negative")
if self.phase_noise_std_rad < 0.0:
    raise ValueError("phase_noise_std_rad cannot be negative")
if self.phase_noise_white_pm_std_rad < 0.0:
    raise ValueError("phase_noise_white_pm_std_rad cannot be negative")
if self.flicker_frequency_std_hz < 0.0:
    raise ValueError("flicker_frequency_std_hz cannot be negative")
if self.shadowing_std_db < 0.0:
    raise ValueError("shadowing_std_db cannot be negative")
if self.shadowing_correlation_s <= 0.0:
    raise ValueError("shadowing_correlation_s must be positive")
if self.adc_bits < 2 or self.dac_bits < 2:
    raise ValueError("ADC and DAC resolution must be at least two bits")
if not 0.0 < self.tx_amplitude <= 1.0:
    raise ValueError("tx_amplitude must be in (0, 1]")
if self.pa_clip_level <= 0.0 or self.agc_target_rms <= 0.0:
    raise ValueError("PA clip level and AGC target must be positive")
if not 0.0 <= self.detection_threshold <= 1.0:
    raise ValueError("detection_threshold must be in [0, 1]")
if self.phase_correction_bits < 1:
    raise ValueError("phase_correction_bits must be positive")
if self.frequency_correction_resolution_hz <= 0.0:
    raise ValueError("frequency_correction_resolution must be positive")
if self.correction_latency_intervals < 0:
    raise ValueError("correction_latency_intervals cannot be negative")
if self.tdd_turnaround_s < 0.0:
    raise ValueError("tdd_turnaround_s cannot be negative")

@property
def sample_period(self) -> float:
    return 1.0 / self.sample_rate
```

Three small data carriers

`SyncPreamble`: the pilot waveform plus the layout facts the receiver needs.
`IQCapture`: what one capture returns – the impaired samples, an impairment-free *oracle* copy (ground truth for diagnostics), and the end value of the intra-frame oscillator walk, which the loop must carry into the clock state. `SDRMeasurement`: what the receiver reports – detected?, phase, frequency, timing, and the detection metric. (`sdr.py:160--191.`)

Three small data carriers — type this

```
@dataclass(frozen=True)
class SyncPreamble:
    waveform: torch.Tensor
    short_sequence: torch.Tensor
    long_sequence: torch.Tensor
    short_length: int
    long_block_length: int

    @property
    def length(self) -> int:
        return self.waveform.numel()
```

```
@dataclass(frozen=True)
class IQCapture:
    samples: torch.Tensor
    oracle_samples: torch.Tensor
    lo_walk_end: torch.Tensor
    inserted_start: int
    expected_arrival: int
    agc_gain: float
    clip_rate: float
```

```
@dataclass(frozen=True)
class SDRMeasurement:
    detected: bool
    phase: torch.Tensor
    frequency: torch.Tensor
    timing_index: int
    detection_metric: float
```

The pilot: Zadoff–Chu training fields

Constant-amplitude chirps with a single-spike autocorrelation – the same family LTE and 5G use. The preamble is 16×16 short repeats (for coarse timing and frequency) then 2047×2 long sequences with a cyclic prefix (for fine frequency and phase). Why 2047: the frequency-estimate error times the 50 ms interval is phase drift the loop must ride out, and it must stay far inside the $\pm\pi/2$ branch boundary you will meet later. (`sdr.py:253--288.`)

The pilot: Zadoff–Chu training fields — type this (1/2)

```
def _zadoff_chu(length: int, root: int, device: torch.device) -> torch.Tensor:
    if math.gcd(length, root) != 1:
        raise ValueError("Zadoff-Chu root and length must be coprime")
    n = torch.arange(length, dtype=REAL_DTYPE, device=device)
    if length % 2:
        phase = -math.pi * root * n * (n + 1.0) / length
    else:
        phase = -math.pi * root * n.square() / length
    return torch.exp(1j * phase).to(COMPLEX_DTYPE)

def make_sync_preamble(
    settings: SDRSimulationConfig, device: torch.device
) -> SyncPreamble:
    """Construct short and long SDR synchronization training fields."""

    short = _zadoff_chu(settings.short_sequence_length, 1, device)
    long_root = 25
    while math.gcd(settings.long_sequence_length, long_root) != 1:
        long_root += 1
    long = _zadoff_chu(settings.long_sequence_length, long_root, device)
    short_field = short.repeat(settings.short_repetitions)
    cyclic_prefix = (
        long[-settings.long_cp_length :] if settings.long_cp_length else long[:0]
    )
    long_block = torch.cat((cyclic_prefix, long))
    long_field = long_block.repeat(settings.long_repetitions)
    waveform = torch.cat((short_field, long_field))
    return SyncPreamble(
        waveform=waveform,
```


The pilot: Zadoff–Chu training fields — type this (2/2)

```
short_sequence=short,  
long_sequence=long,  
short_length=short_field.numel(),  
long_block_length=long_block.numel(),  
)
```

Converter and clock imperfections

`_quantize_iq`: a b -bit converter – clip to full scale, round to the grid, report the clip rate. `_soft_limit`: the power amplifier's gentle saturation.
`_resample_clock_offset`: the receiver's sample clock runs at a slightly different rate than the transmitter's, so the samples are linearly re-interpolated at the offset rate plus a fractional delay. (`sdr.py`:290--325.)

Converter and clock imperfections — type this (1/2)

```
def _quantize_iq(samples: torch.Tensor, bits: int) -> tuple[torch.Tensor, float]:
    limit = float(2 ** (bits - 1) - 1)
    real = samples.real
    imag = samples.imag
    clipped = (real.abs() > 1.0) | (imag.abs() > 1.0)
    real = torch.round(torch.clamp(real, -1.0, 1.0) * limit) / limit
    imag = torch.round(torch.clamp(imag, -1.0, 1.0) * limit) / limit
    return torch.complex(real, imag), torch.mean(clipped.to(torch.float)).item()

def _soft_limit(samples: torch.Tensor, limit: float) -> torch.Tensor:
    magnitude = torch.abs(samples)
    scale = torch.clamp(limit / torch.clamp(magnitude, min=1e-15, max=1.0), min=1e-15, max=1.0)
    return samples * scale

def _resample_clock_offset(
    samples: torch.Tensor, ppm: float, delay_samples: float = 0.0
) -> torch.Tensor:
    """Linearly resample IQ at a receiver clock offset and fractional delay."""
```

Converter and clock imperfections — type this (2/2)

```
if ppm == 0.0 and delay_samples == 0.0:
    return samples
ratio = 1.0 + ppm * 1e-6
output_index = torch.arange(
    samples.numel(), dtype=REAL_DTYPE, device=samples.device
)
source = output_index / ratio - delay_samples
lower = torch.floor(source).to(torch.int64)
upper = lower + 1
valid = (lower >= 0) & (upper < samples.numel())
lower = torch.clamp(lower, 0, samples.numel() - 1)
upper = torch.clamp(upper, 0, samples.numel() - 1)
fraction = source - lower.to(REAL_DTYPE)
output = samples[lower] * (1.0 - fraction) + samples[upper] * fraction
return torch.where(valid, output, torch.zeros_like(output))
```

Flicker frequency noise

Real oscillators have $1/f$ frequency noise on top of the white walks. The standard time-domain surrogate: a bank of four first-order decaying processes with log-spaced time constants, summed. The class also exposes its per-interval `innovation_variance` – the loop adds it to the filter's process noise, so the filter knows about this noise source too. (`sdr.py:328--391.`)

Flicker frequency noise — type this (1/2)

```
class _FlickerFrequencyNoise:
    """Approximate flicker FM with a bank of first-order Gauss-Markov processes.

    A sum of AR(1) components with log-spaced correlation times approximates a
    1/f frequency-noise spectrum across the timescales the simulation spans,
    which is the standard time-domain surrogate for flicker FM.
    """

    def __init__(
        self,
        std_hz: float,
        interval_s: float,
        horizon_s: float,
        device: torch.device,
        generator: torch.Generator,
        components: int = 4,
    ) -> None:
        self.device = device
        self.generator = generator
        self.enabled = std_hz > 0.0
        if not self.enabled:
            self.innovation_variance = 0.0
            return
        shortest = 2.0 * interval_s
        longest = max(horizon_s, 4.0 * interval_s)
        exponents = torch.linspace(
            math.log10(shortest),
            math.log10(longest),
            components,
            dtype=REAL_DTYPE,
```

Flicker frequency noise — type this (2/2)

```
        device=device,
    )
    correlation_times = torch.pow(
        torch.tensor(10.0, dtype=REAL_DTYPE, device=device), exponents
    )
    self.rho = torch.exp(-interval_s / correlation_times)
    component_std = 2.0 * math.pi * std_hz / math.sqrt(components)
    self.sigma = torch.full(
        (components,), component_std, dtype=REAL_DTYPE, device=device
    )
    self.state = self.sigma * torch.randn(
        components, dtype=REAL_DTYPE, device=device, generator=generator
    )
    self.innovation_variance = float(
        torch.sum(2.0 * self.sigma.square() * (1.0 - self.rho)).item()
    )

def step(self) -> torch.Tensor:
    """Advance one interval; return the flicker frequency offset in rad/s."""

    if not self.enabled:
        return torch.zeros((), dtype=REAL_DTYPE, device=self.device)
    innovation = (
        torch.randn(
            self.sigma.numel(),
            dtype=REAL_DTYPE,
            device=self.device,
            generator=self.generator,
        )
        * self.sigma
        * torch.sqrt(1.0 - self.rho.square())
    )
    self.state = self.rho * self.state + innovation
    return torch.sum(self.state)
```

SDRRadioLink: the channel

The propagation channel is Sionna's 3GPP tapped-delay-line, converted to a 16-tap discrete-time filter, one realization per sync interval. **mirror_of is reciprocity in one argument**: the reverse link of a two-way pair shares the forward link's taps and shadowing – the physical fact the entire method stands on. The noise floor is fixed at the nominal channel gain, so fading changes the actual signal-to-noise ratio, as in the field. (`sdr.py:394--473.`)

SDRRadioLink: the channel — type this (1/3)

```
class SDRRadioLink:
    """Sampled-IQ transmit/channel/receive chain using Sionna TDL and AWGN."""

    def __init__(
        self,
        settings: SDRSimulationConfig,
        preamble: SyncPreamble,
        device: torch.device,
        generator: torch.Generator,
        mirror_of: "SDRRadioLink | None" = None,
        captures_per_interval: int = 1,
    ) -> None:
        self.settings = settings
        self.preamble = preamble
        self.device = device
        self.generator = generator
        self.captures_per_interval = captures_per_interval
        # A mirrored link models the reverse direction of a reciprocal
        # channel: it shares the taps and shadowing of the forward link (which
        # must capture first each interval) while keeping its own receiver
        # state (timing carry, noise floor).
        self._mirror = mirror_of
        self._last_shadow_amplitude = 1.0
        self.input_length = (
            preamble.length
            + 2 * settings.capture_guard_samples
            + settings.timing_jitter_samples
        )
        self.l_min, self.l_max = time_lag_discrete_time_channel(
            settings.sample_rate, settings.maximum_channel_delay_s
```

SDRRadioLink: the channel — type this (2/3)

```
)
self.l_tot = self.l_max - self.l_min + 1
self.interval_samples = int(
    round(settings.sync_interval * settings.sample_rate)
)
# Receiver clock error accumulated since the capture window was last
# re-centered, in samples.
self._timing_carry = 0.0
# Thermal noise floor, fixed after the first frame at the nominal
# (unshadowed) channel gain so fading changes the actual SNR.
self._noise_power: torch.Tensor | None = None
self._shadow_rho = math.exp(
    -settings.sync_interval / settings.shadowing_correlation_s
)
self._shadow_db = settings.shadowing_std_db * torch.randn(
    (), dtype=REAL_DTYPE, device=device, generator=generator
)

if mirror_of is not None:
    self.channel_taps = mirror_of.channel_taps
else:
    self.tdl = TDL(
        model=settings.tdl_model,
        delay_spread=settings.delay_spread_s,
        carrier_frequency=settings.carrier_frequency_hz,
        min_speed=settings.channel_speed_mps,
        max_speed=settings.channel_speed_mps,
        precision="double",
        device=str(device),
    )
```

SDRRadioLink: the channel — type this (3/3)

```
coefficients, delays = self.tdl(
    batch_size=1,
    num_time_steps=settings.num_iterations,
    sampling_frequency=1.0 / settings.sync_interval,
)
self.channel_taps = cir_to_time_channel(
    settings.sample_rate,
    coefficients,
    delays,
    self.l_min,
    self.l_max,
    normalize=True,
)
self._apply_channel = ApplyTimeChannel(
    num_time_samples=self.input_length,
    l_tot=self.l_tot,
    precision="double",
    device=str(device),
)
self._awgn = AWGN(precision="double", device=str(device))
```

Link helpers: jitter and shadowing

`_random_start`: the capture window lands with random timing jitter.
`_channel_for_frame`: this interval's taps, shaped for Sionna's channel operator.
`_step_shadowing`: slow log-normal fading as a correlated process – and the mirrored link reuses the forward link's draw (reciprocity again).
(`sdr.py`:475--508.)

Link helpers: jitter and shadowing — type this

```
def _random_start(self) -> int:
    jitter = torch.randint(
        self.settings.timing_jitter_samples + 1,
        (),
        device=self.device,
        generator=self.generator,
    ).item()
    return self.settings.capture_guard_samples + jitter

def _channel_for_frame(self, iteration: int) -> torch.Tensor:
    output_length = self.input_length + self.l_tot - 1
    taps = self.channel_taps[..., iteration, :].unsqueeze(-2)
    return taps.expand(*taps.shape[:-2], output_length, self.l_tot)

def _step_shadowing(self) -> float:
    """Advance the temporally correlated log-normal shadowing process."""

    if self._mirror is not None:
        # Reciprocity: reuse the forward link's shadowing for this frame.
        return self._mirror._last_shadow_amplitude
    if self.settings.shadowing_std_db <= 0.0:
        self._last_shadow_amplitude = 1.0
        return 1.0
    innovation = torch.randn(
        (), dtype=REAL_DTYPE, device=self.device, generator=self.generator
    )
    self._shadow_db = (
        self._shadow_rho * self._shadow_db
        + math.sqrt(1.0 - self._shadow_rho**2)
        * self.settings.shadowing_std_db
        * innovation
    )
    self._last_shadow_amplitude = float(10.0 ** (self._shadow_db.item() / 20.0))
    return self._last_shadow_amplitude
```

capture(), transmit side

One direction of one exchange, in physical order: the receiver's sample-clock error moves the arrival window (whole samples step it; the sub-sample residue stays as a fractional delay); the transmitter imprints *its* oscillator on the waveform; the power amplifier soft-clips; the DAC quantizes; Sionna's channel convolves; shadowing scales. (`sdr.py:510--544.`)

capture(), transmit side — type this (1/2)

```
def capture(
    self, master: Oscillator, slave: Oscillator, iteration: int, sfo_ppm: float
) -> IQCapture:
    settings = self.settings
    # Clock error accumulates between frames. The receiver re-centers its
    # window on each detection, so whole-sample drift steps the insertion
    # point while the sub-sample residual remains as a fractional delay.
    self._timing_carry += (
        sfo_ppm * 1e-6 * self.interval_samples / self.captures_per_interval
    )
    drift_step = int(round(self._timing_carry))
    self._timing_carry -= drift_step
    fractional_delay = self._timing_carry
    start = self._random_start() + drift_step
    start = min(max(start, 0), self.input_length - self.preamble.length)
    center = start + (self.preamble.length - 1) / 2.0
    sample_index = torch.arange(
        self.input_length, dtype=REAL_DTYPE, device=self.device
    )
    time = (sample_index - center) * settings.sample_period

    frame = torch.zeros(self.input_length, dtype=COMPLEX_DTYPE, device=self.device)
    frame[start : start + self.preamble.length] = (
        settings.tx_amplitude * self.preamble.waveform
    )
    master_carrier = torch.exp(1j * (master.phase + master.frequency * time))
    tx = _soft_limit(frame * master_carrier, settings.pa_clip_level)
    tx, _ = _quantize_iq(tx, settings.dac_bits)
```

capture(), transmit side — type this (2/2)

```
channel_input = tx.reshape(1, 1, 1, -1)
clean = self._apply_channel(
    channel_input, self._channel_for_frame(iteration)
).reshape(-1)
shadow_amplitude = self._step_shadowing()
clean = clean * shadow_amplitude
```


capture(), receive side

Thermal noise at the fixed floor; sample-clock resampling; down-conversion with the *receiver's* oscillator (what remains is the clock difference plus the channel); the oscillator's own random walk *during* the frame – whose end value is returned so the loop keeps the noise continuous across frames; white phase jitter; IQ imbalance; automatic gain control; DC offset; 12-bit quantization. Every phase error the filter must fight is created here, physically. (`sdr.py:545--618.`)

capture(), receive side — type this (1/3)

```
active_start = max(0, start - self.l_min)
active_stop = min(clean.numel(), active_start + self.preamble.length)
if self._noise_power is None:
    nominal_power = torch.mean(
        torch.abs(clean[active_start:active_stop] / shadow_amplitude).square()
    )
    self._noise_power = nominal_power / (10.0 ** (settings.snr_db / 10.0))
received = self._awgn(clean, self._noise_power)

received = _resample_clock_offset(received, sfo_ppm, fractional_delay)
# The oracle path shares the frame, channel realization, shadowing,
# deterministic transmit hardware, and sample-clock timing, but carries
# no stochastic noise or receiver impairments. It defines the
# ground-truth observable.
oracle = _resample_clock_offset(clean, sfo_ppm, fractional_delay)
output_index = torch.arange(
    received.numel(), dtype=REAL_DTYPE, device=self.device
)
physical_index = output_index / (1.0 + sfo_ppm * 1e-6) + self.l_min
receive_time = (physical_index - center) * settings.sample_period
slave_carrier = torch.exp(-1j * (slave.phase + slave.frequency * receive_time))
received = received * slave_carrier
oracle = oracle * slave_carrier
```

capture(), receive side — type this (2/3)

```
lo_walk_end = torch.zeros(), dtype=REAL_DTYPE, device=self.device)
if settings.phase_noise_std_rad > 0.0:
    phase_noise = torch.cumsum(
        torch.randn(
            received.numel(),
            dtype=REAL_DTYPE,
            device=self.device,
            generator=self.generator,
        )
        * settings.phase_noise_std_rad,
        dim=0,
    )
    received = received * torch.exp(1j * phase_noise)
    # End value of the intra-frame walk; the caller carries it into
    # the oscillator state so the LO noise is one continuous process.
    lo_walk_end = phase_noise[-1]

if settings.phase_noise_white_pm_std_rad > 0.0:
    jitter = (
        torch.randn(
            received.numel(),
            dtype=REAL_DTYPE,
            device=self.device,
            generator=self.generator,
        )
        * settings.phase_noise_white_pm_std_rad
    )
    received = received * torch.exp(1j * jitter)
```

capture(), receive side — type this (3/3)

```
gain_ratio = 10.0 ** (settings.iq_gain_imbalance_db / 40.0)
phase_skew = math.radians(settings.iq_phase_imbalance_deg)
in_phase = received.real * gain_ratio
quadrature = received.imag / gain_ratio
quadrature = quadrature * math.cos(phase_skew) + in_phase * math.sin(phase_skew)
received = torch.complex(in_phase, quadrature)

rms = torch.sqrt(torch.mean(torch.abs(received).square()))
agc_gain = torch.clamp(settings.agc_target_rms / rms, 0.01, 1000.0)
received = received * agc_gain + settings.dc_offset
received, clip_rate = _quantize_iq(received, settings.adc_bits)

return IQCapture(
    samples=received,
    oracle_samples=oracle,
    lo_walk_end=lo_walk_end,
    inserted_start=start,
    expected_arrival=start - self.l_min,
    agc_gain=agc_gain.item(),
    clip_rate=clip_rate,
)
```

The receiver, stage 1: coarse timing and frequency

The receiver knows neither when the pilot arrives nor the frequency offset. Correlating the signal with itself 16 samples later solves both at once: when the repeating short field fills the window, consecutive repeats match – the normalized magnitude peaks (timing), and the angle between repeats is the offset accumulated over 16 samples (frequency, unambiguous to $\pm f_s/32 = \pm 31$ kHz). (sdr.py:621--659.)

The receiver, stage 1: coarse timing and frequency — type this (1/2)

```
def _sliding_sum(values: torch.Tensor, width: int) -> torch.Tensor:
    cumulative = torch.cat(
        (
            torch.zeros(1, dtype=values.dtype, device=values.device),
            torch.cumsum(values, dim=0),
        )
    )
    return cumulative[width:] - cumulative[:-width]

class SDRSynchronizer:
    """Packet detector and two-stage carrier synchronizer."""

    def __init__(
        self,
        settings: SDRSimulationConfig,
        preamble: SyncPreamble,
    ) -> None:
        self.settings = settings
        self.preamble = preamble
```

The receiver, stage 1: coarse timing and frequency — type this (2/2)

```
def _coarse_timing_and_frequency(
    self, samples: torch.Tensor
) -> tuple[int, torch.Tensor]:
    lag = self.settings.short_sequence_length
    width = self.preamble.short_length - lag
    product = torch.conj(samples[:-lag]) * samples[lag:]
    first_power = torch.abs(samples[:-lag]).square()
    second_power = torch.abs(samples[lag:]).square()
    correlation = _sliding_sum(product, width)
    normalization = torch.sqrt(
        _sliding_sum(first_power, width) * _sliding_sum(second_power, width)
    ).clamp_min(1e-15)
    metric = torch.abs(correlation) / normalization
    coarse_start = int(torch.argmax(metric).item())
    frequency = torch.angle(correlation[coarse_start]) / (
        lag * self.settings.sample_period
    )
    return coarse_start, frequency
```

The receiver, stages 2–3: detect, refine, read the phase

Stage 2: after coarse derotation, the *whole* preamble is matched against every window; the normalized peak gives precise timing, and its height is the detection metric – below the 0.25 threshold the capture is declared failed and the loop must coast. Stage 3: the two long fields are identical and 2175 samples apart; the angle between them refines the frequency to ~ 0.1 Hz. Finally, derotate with the refined frequency (indices centered so the phase is referenced to frame *center*) and matched-filter the long fields: the correlation angle is the carrier phase – clock offset **plus the channel's phase**. Separating those two is the loop's job. (sdr.py:661--754.)

The receiver, stages 2–3: detect, refine, read the phase — type this (1/3)

```
def estimate(self, samples: torch.Tensor) -> SDRMeasurement:
    samples = samples - torch.mean(samples)
    coarse_start, coarse_frequency = self._coarse_timing_and_frequency(samples)

    sample_index = torch.arange(
        samples.numel(), dtype=REAL_DTYPE, device=samples.device
    )
    coarse_corrected = samples * torch.exp(
        -1j
        * coarse_frequency
        * (sample_index - coarse_start)
        * self.settings.sample_period
    )

    if coarse_corrected.numel() < self.preamble.length:
        return SDRMeasurement(
            False,
            torch.zeros(), dtype=REAL_DTYPE, device=samples.device),
            coarse_frequency,
            coarse_start,
            0.0,
        )

    windows = coarse_corrected.unfold(0, self.preamble.length, 1)
    correlations = torch.sum(windows * torch.conj(self.preamble.waveform), dim=-1)
    window_energy = torch.sum(torch.abs(windows).square(), dim=-1)
    reference_energy = torch.sum(torch.abs(self.preamble.waveform).square())
    normalized = torch.abs(correlations) / torch.sqrt(
        (window_energy * reference_energy).clamp_min(1e-15)
    )
    timing = int(torch.argmax(normalized).item())
```

The receiver, stages 2–3: detect, refine, read the phase — type this (2/3)

```
detection_metric = normalized[timing].item()

short = samples[timing : timing + self.preamble.short_length]
lag = self.settings.short_sequence_length
short_correlation = torch.sum(torch.conj(short[:-lag]) * short[lag:])
coarse_frequency = torch.angle(short_correlation) / (
    lag * self.settings.sample_period
)

segment = samples[timing : timing + self.preamble.length]
relative_index = torch.arange(
    segment.numel(), dtype=REAL_DTYPE, device=samples.device
)
coarse_segment = segment * torch.exp(
    -1j * coarse_frequency * relative_index * self.settings.sample_period
)

first_start = self.preamble.short_length + self.settings.long_cp_length
last_start = (
    self.preamble.short_length
    + (self.settings.long_repetitions - 1) * self.preamble.long_block_length
    + self.settings.long_cp_length
)
first_long = coarse_segment[
    first_start : first_start + self.settings.long_sequence_length
]
last_long = coarse_segment[
    last_start : last_start + self.settings.long_sequence_length
]
separation = (
```

The receiver, stages 2–3: detect, refine, read the phase — type this (3/3)

```
        self.settings.long_repetitions - 1
    ) * self.preamble.long_block_length
fine_correlation = torch.sum(torch.conj(first_long) * last_long)
fine_frequency = torch.angle(fine_correlation) / (
    separation * self.settings.sample_period
)
frequency = coarse_frequency + fine_frequency

centered_index = relative_index - (self.preamble.length - 1) / 2.0
corrected = segment * torch.exp(
    -1j * frequency * centered_index * self.settings.sample_period
)
phase_correlation = torch.zeros(), dtype=COMPLEX_DTYPE, device=samples.device)
for repeat in range(self.settings.long_repetitions):
    sequence_start = (
        self.preamble.short_length
        + repeat * self.preamble.long_block_length
        + self.settings.long_cp_length
    )
    sequence = corrected[
        sequence_start : sequence_start + self.settings.long_sequence_length
    ]
    phase_correlation += torch.sum(
        torch.conj(self.preamble.long_sequence) * sequence
    )
phase = torch.angle(phase_correlation)
detected = detection_metric >= self.settings.detection_threshold
return SDRMeasurement(
    detected=detected,
    phase=phase,
    frequency=frequency,
    timing_index=timing,
    detection_metric=detection_metric,
)
```

R from physics, and quantized corrections

The filter's measurement covariance is **derived, not tuned**: the textbook estimation bound for a pilot of this length and signal-to-noise ratio, plus the phase noise the oscillator smears across the frame, plus per-sample jitter. This is why no gain is hand-tuned anywhere. `_quantize_correction` snaps commands to what hardware can apply: 16-bit phase steps, 0.01 Hz frequency steps.
(`sdr.py:757--815.`)

R from physics, and quantized corrections — type this (1/2)

```
def _measurement_covariance(
    settings: SDRSimulationConfig, preamble: SyncPreamble, device: torch.device
) -> torch.Tensor:
    snr = 10.0 ** (settings.snr_db / 10.0)
    phase_variance = 1.0 / (
        2.0 * snr * settings.long_sequence_length * settings.long_repetitions
    )
    separation_time = (
        (settings.long_repetitions - 1)
        * preamble.long_block_length
        * settings.sample_period
    )
    frequency_variance = 1.0 / (
        snr * settings.long_sequence_length * separation_time**2
    )
    # The receiver LO phase noise accumulates as a random walk across the
    # frame and usually dominates the AWGN Cramer-Rao bound. The phase
    # estimate averages the walk over the long training fields, and the fine
    # CFO estimate differences the walk between the two fields.
    long_field_offset = preamble.short_length + settings.long_cp_length
    long_field_span = preamble.length - long_field_offset
    phase_noise_variance = settings.phase_noise_std_rad**2
    phase_variance += phase_noise_variance * (
        long_field_offset + long_field_span / 3.0
    )
    separation_samples = (
        settings.long_repetitions - 1
    ) * preamble.long_block_length
    frequency_variance += (
        phase_noise_variance * separation_samples / separation_time**2
    )
```

R from physics, and quantized corrections — type this (2/2)

```
)
# White PM adds an independent phase jitter per sample; it averages down
# over the long training fields but still belongs in the noise budget.
white_pm_variance = settings.phase_noise_white_pm_std_rad**2
total_long_samples = settings.long_sequence_length * settings.long_repetitions
phase_variance += white_pm_variance / total_long_samples
frequency_variance += 2.0 * white_pm_variance / (
    settings.long_sequence_length * separation_time**2
)
return torch.diag(
    torch.tensor(
        [phase_variance, phase_variance, frequency_variance],
        dtype=REAL_DTYPE,
        device=device,
    )
)

def _quantize_correction(
    correction: torch.Tensor, settings: SDRSimulationConfig
) -> torch.Tensor:
    phase_step = 2.0 * math.pi / (2**settings.phase_correction_bits)
    frequency_step = 2.0 * math.pi * settings.frequency_correction_resolution_hz
    return torch.stack(
        (
            torch.round(correction[0] / phase_step) * phase_step,
            torch.round(correction[1] / frequency_step) * frequency_step,
        )
    )
```

The extended Kalman filter: state, predict, and the circle

The filter tracks the *relative* state $[\hat{\theta}, \hat{\omega}]$ with the same transition matrix as the Oscillator. `predict`: push the estimate through the drift model, inflate the covariance – “the clocks kept drifting while I wasn’t looking.” The measurement is a **point on the unit circle**, $[\cos \phi, \sin \phi, \omega]$, so no innovation ever wraps; the price is a nonlinear h , linearized by the Jacobian – which is the entire meaning of “extended.” (`core.py`:286--339.)

The extended Kalman filter: state, predict, and the circle — type this (1/2)

```
class PhaseFrequencyEKF:
    """Iterated EKF for relative phase and angular-frequency state."""

    def __init__(
        self,
        sync_interval: float,
        process_covariance: torch.Tensor,
        measurement_covariance_matrix: torch.Tensor,
        device: torch.device,
        initial_covariance: Optional[torch.Tensor] = None,
    ) -> None:
        self.device = device
        self.state = torch.zeros(2, dtype=REAL_DTYPE, device=device)
        if initial_covariance is None:
            self.covariance = torch.eye(2, dtype=REAL_DTYPE, device=device)
        else:
            self.covariance = _as_matrix(initial_covariance, device)
            if self.covariance.shape != (2, 2):
                raise ValueError("initial_covariance must have shape [2, 2]")
        self.transition = torch.tensor(
            [[1.0, sync_interval], [0.0, 1.0]],
            dtype=REAL_DTYPE,
            device=device,
        )
        self.process_covariance = _as_matrix(process_covariance, device)
        self.measurement_covariance = _as_matrix(measurement_covariance_matrix, device)
        if self.process_covariance.shape != (2, 2):
            raise ValueError("process_covariance must have shape [2, 2]")
        if self.measurement_covariance.shape != (3, 3):
            raise ValueError("measurement covariance must have shape [3, 3]")
```


The extended Kalman filter: state, predict, and the circle — type this (2/2)

```
def predict(self) -> None:
    self.state = self.transition @ self.state
    self.covariance = (
        self.transition @ self.covariance @ self.transition.T
        + self.process_covariance
    )

def _measurement(self) -> torch.Tensor:
    return torch.stack(
        (torch.cos(self.state[0]), torch.sin(self.state[0]), self.state[1])
    )

def _jacobian(self) -> torch.Tensor:
    phi = self.state[0]
    zero = torch.zeros((), dtype=REAL_DTYPE, device=self.device)
    one = torch.ones((), dtype=REAL_DTYPE, device=self.device)
    return torch.stack(
        (
            torch.stack((-torch.sin(phi), zero)),
            torch.stack((torch.cos(phi), zero)),
            torch.stack((zero, one)),
        )
    )
```

The iterated update, Joseph form, and the reset

Three deliberate choices. **Iterate**: with a large offset one linearized shot under-corrects (the comment says it best: $\sin(-1.4)$ is not ≈ -1.4) – so update, re-linearize, repeat until converged. **Joseph form**: the textbook covariance shortcut can go asymmetric or negative under rounding; this form cannot. **reset_after_correction**: after commanding the clock to shift, subtract the command from the estimate – the filter tracks *what remains*. Skip it and every correction applies twice. (`core.py:341--382.`)

The iterated update, Joseph form, and the reset — type this (1/2)

```
def update(self, measurement: torch.Tensor) -> None:
    measurement = measurement.to(dtype=REAL_DTYPE, device=self.device)
    if measurement.shape != (3,):
        raise ValueError("measurement must have shape [3]")

    prior_state = self.state.clone()
    prior_covariance = self.covariance.clone()
    posterior_state = prior_state
    # Re-linearization avoids the large-offset under-correction of a
    # one-shot EKF update (e.g., sin(-1.4) is not approximately -1.4).
    for _ in range(6):
        self.state = posterior_state
        jacobian = self._jacobian()
        innovation_covariance = (
            jacobian @ prior_covariance @ jacobian.T + self.measurement_covariance
        )
        kalman_gain = torch.linalg.solve(
            innovation_covariance,
            jacobian @ prior_covariance,
        ).T
        iterated_innovation = (
            measurement
            - self._measurement()
            - jacobian @ (prior_state - posterior_state)
        )
        next_state = prior_state + kalman_gain @ iterated_innovation
        if torch.max(torch.abs(next_state - posterior_state)) < 1e-10:
            posterior_state = next_state
            break
    posterior_state = next_state
```

The iterated update, Joseph form, and the reset — type this (2/2)

```
self.state = posterior_state

identity = torch.eye(2, dtype=REAL_DTYPE, device=self.device)
residual_map = identity - kalman_gain @ jacobian
# Joseph form keeps the covariance symmetric and positive semidefinite.
self.covariance = (
    residual_map @ prior_covariance @ residual_map.T
    + kalman_gain @ self.measurement_covariance @ kalman_gain.T
)

def reset_after_correction(self, correction: torch.Tensor) -> None:
    self.state = self.state - correction
```

The result object

Everything the run records, with the honest masks built in:
`steady_state_phase_rms` and `mean_coherent_gain` only count intervals where
detected & `correction_active` & calibrated – steady state means *after*
acquisition, actuation, and the π check, and the code says so explicitly.
(`coherent.py:42--101.`)

The result object — type this (1/2)

```
@dataclass(frozen=True)
class TwoWaySimulationResult:
    """Metrics from a reciprocal two-way synchronization run.

    Unlike the one-way loop, the oscillator offset is identifiable, so all
    phase metrics are referenced to the true relative oscillator state.
    """

    true_phase: torch.Tensor
    measured_phase: torch.Tensor
    estimated_phase: torch.Tensor
    true_frequency: torch.Tensor
    physical_relative_frequency: torch.Tensor
    estimated_frequency: torch.Tensor
    phase_error: torch.Tensor
    frequency_error: torch.Tensor
    post_correction_phase: torch.Tensor
    post_correction_frequency: torch.Tensor
    coherent_gain: torch.Tensor
    detected: torch.Tensor
    correction_active: torch.Tensor
    calibrated: torch.Tensor
    airtime_fraction: float
    device: torch.device

    @property
    def detection_rate(self) -> float:
        return torch.mean(self.detected.to(torch.float64)).item()
```

The result object — type this (2/2)

```
@property
def phase_rmse(self) -> float:
    valid = self.detected
    if not torch.any(valid):
        return float("nan")
    return torch.sqrt(torch.mean(self.phase_error[valid].square())).item()

@property
def steady_state_phase_rms(self) -> float:
    valid = self.detected & self.correction_active & self.calibrated
    if not torch.any(valid):
        return float("nan")
    return torch.sqrt(
        torch.mean(self.post_correction_phase[valid].square())
    ).item()

@property
def mean_coherent_gain(self) -> float:
    valid = self.detected & self.correction_active & self.calibrated
    if not torch.any(valid):
        return float("nan")
    return torch.mean(self.coherent_gain[valid]).item()

@property
def final_phase_error(self) -> float:
    return self.post_correction_phase[-1].item()

@property
def final_frequency_error_hz(self) -> float:
    return self.post_correction_frequency[-1].item() / (2.0 * math.pi)
```

`_pick_half_phase`: the two candidate answers

The two-way trick divides a phase difference by two, and division by two on a circle loses one bit: the true offset is either the measurement or the measurement $+\pi$ – *exactly* two candidates, at any signal-to-noise ratio. Once locked, the offset cannot physically jump by π in one interval, so the filter's own prediction picks the branch. At first contact there is no prediction – that hole is plugged by the one-time power check inside the loop. (`coherent.py:104--119`.)

`_pick_half_phase`: the two candidate answers — type this

```
def _pick_half_phase(
    combined_half: torch.Tensor, reference: torch.Tensor
) -> torch.Tensor:
    """Resolve the pi ambiguity of a half-difference phase.

    (a - b)/2 is only defined modulo pi. The candidate closer to the
    reference (the EKF prediction after acquisition) is chosen; the global
    pi ambiguity at acquisition is a real property of two-way carrier
    alignment and is resolved in hardware by a one-time combining check.
    """

    alternate = wrap_phase(combined_half + math.pi)
    keep = torch.abs(wrap_phase(combined_half - reference)) <= torch.abs(
        wrap_phase(alternate - reference)
    )
    return torch.where(keep, combined_half, alternate)
```

run_two_way_simulation: setup

Seeds (note `sionna_config.seed` – the channel is pinned by Sionna's own seed, not torch's; a real trap), two oscillators sharing one covariance, the reverse link as a `mirror_of` the forward link, and the filter's matrices assembled from physics: R = the derived covariance *halved* (the half-difference averages two independent captures); Q = *twice* the per-oscillator covariance (two independent clocks make the relative state) + the white-FM walk over one interval + the flicker innovation. Initial covariance: phase $\pm\pi$, frequency ± 50 kHz – honest ignorance. Airtime is accounted exactly: two captures per interval. (`coherent.py`:122--231.)

run_two_way_simulation: setup — type this (1/4)

```
def run_two_way_simulation(
    settings: SDRSimulationConfig = SDRSimulationConfig(),
) -> TwoWaySimulationResult:
    """Run reciprocal two-way OTA synchronization for open-loop coherence."""

    device = resolve_device(settings.device)
    torch.manual_seed(settings.seed)
    if device.type == "cuda":
        torch.cuda.manual_seed_all(settings.seed)
    sionna_config.seed = settings.seed
    generator = torch.Generator(device=device)
    generator.manual_seed(settings.seed + 1)

    frequency_process_std = 2.0 * math.pi * settings.frequency_process_std_hz
    oscillator_covariance = torch.diag(
        torch.tensor(
            [settings.phase_process_std_rad**2, frequency_process_std**2],
            dtype=REAL_DTYPE,
            device=device,
        )
    )
    master = Oscillator(
        settings.master_initial_phase,
        2.0 * math.pi * settings.master_initial_frequency_hz,
        settings.sync_interval,
        oscillator_covariance,
        device,
        generator,
    )
    slave = Oscillator(
```

run_two_way_simulation: setup — type this (2/4)

```
settings.slave_initial_phase,  
2.0 * math.pi * settings.slave_initial_frequency_hz,  
settings.sync_interval,  
oscillator_covariance,  
device,  
generator,  
)  
preamble = make_sync_preamble(settings, device)  
link_forward = SDRRadioLink(settings, preamble, device, generator)  
link_reverse = SDRRadioLink(  
    settings, preamble, device, generator, mirror_of=link_forward  
)  
synchronizer = SDRSynchronizer(settings, preamble)  
# Averaging two independent directional measurements halves the noise.  
measurement_noise = 0.5 * _measurement_covariance(settings, preamble, device)  
  
interval_samples = int(round(settings.sync_interval * settings.sample_rate))  
white_fm_phase_variance = settings.phase_noise_std_rad**2 * interval_samples  
flicker = _FlickerFrequencyNoise(  
    settings.flicker_frequency_std_hz,  
    settings.sync_interval,  
    settings.num_iterations * settings.sync_interval,  
    device,  
    generator,  
)  
ekf = PhaseFrequencyEKF(  
    settings.sync_interval,  
    2.0 * oscillator_covariance  
    + torch.diag(  
        torch.tensor(  

```

run_two_way_simulation: setup — type this (3/4)

```
[white_fm_phase_variance, flicker.innovation_variance],
dtype=REAL_DTYPE,
device=device,
    )
),
measurement_noise,
device,
initial_covariance=torch.diag(
    torch.tensor(
        [math.pi**2, (2.0 * math.pi * 50e3) ** 2],
        dtype=REAL_DTYPE,
        device=device,
    )
),
)

chain_bias = math.radians(settings.twoway_chain_asymmetry_deg)
capture_samples = link_forward.input_length + link_forward.l_tot - 1
remainder_samples = max(0, interval_samples - 2 * capture_samples)
airtime_fraction = (
    2.0 * capture_samples / (settings.sync_interval * settings.sample_rate)
)

pending_corrections: dict[int, torch.Tensor] = {}
carried_lo_walk = torch.zeros(), dtype=REAL_DTYPE, device=device)
flicker_previous = torch.zeros(), dtype=REAL_DTYPE, device=device)
slave_frequency_corrections = torch.zeros(), dtype=REAL_DTYPE, device=device)
correction_has_loaded = False
acquired = False
settled_corrections = 0
pi_calibrated = False
```

run_two_way_simulation: setup — type this (4/4)

```
history: dict[str, list[torch.Tensor]] = {
    name: []
    for name in (
        "true_phase",
        "measured_phase",
        "estimated_phase",
        "true_frequency",
        "physical_relative_frequency",
        "estimated_frequency",
        "phase_error",
        "frequency_error",
        "post_correction_phase",
        "post_correction_frequency",
        "coherent_gain",
        "detected",
        "correction_active",
        "calibrated",
    )
}
```

The loop: drift, delayed corrections, the π check

Each interval starts with physics: both clocks walk; the master absorbs last frame's carried oscillator walk and the flicker step. A correction scheduled `correction_latency_intervals` ago lands now. Then the **one-time π calibration**: the acquisition branch was a guess (wrong about half the time, with the pair transmitting in opposite phase while every signal reads “locked”); three settled corrections after lock, one coarse combined-power check – destructive $\Rightarrow$ flip by π , once. The comment explains why the filter needs no reset: the measurement cannot see a π shift. (`coherent.py:233--264.`)

The loop: drift, delayed corrections, the pi check — type this

```
for iteration in range(settings.num_iterations):
    master.step()
    slave.step()
    master.state[0] = wrap_phase(master.state[0] + carried_lo_walk)
    flicker_now = flicker.step()
    master.state[1] = master.state[1] + (flicker_now - flicker_previous)
    flicker_previous = flicker_now

    due_correction = pending_corrections.pop(iteration, None)
    if due_correction is not None:
        slave.apply_correction(due_correction)
        slave_frequency_corrections = (
            slave_frequency_corrections + due_correction[1]
        )
        correction_has_loaded = True

    # One-time pi-ambiguity calibration: a deployment checks once after
    # lock whether the pair combines destructively (a single coarse
    # power measurement) and flips the NCO by pi if so.
    if correction_has_loaded and not pi_calibrated:
        settled_corrections += 1
        if settled_corrections >= 3:
            if torch.cos(master.state[0] - slave.state[0]) < 0.0:
                # The half-difference measurement is invariant under pi
                # shifts, so the filter needs no reset: its near-zero
                # state re-attaches to the true branch after the flip.
                slave.apply_correction(
                    torch.tensor(
                        [math.pi, 0.0], dtype=REAL_DTYPE, device=device
                    )
                )
            pi_calibrated = True
```


The loop: two captures and the gap between them

Bookkeeping first: the correction-free “physical” clock (what hardware would do with no synchronization at all) yields the sample-clock offset for the captures. Then the forward capture; its intra-frame oscillator walk joins the true state *before* the reverse frame fires – one continuous noise process. Between directions sits a real 1 ms transmit/receive turnaround: both clocks advance at their current frequencies and pick up random walk. Then the reverse capture, and the leftover walk for the rest of the interval is drawn and carried to the next iteration. (`coherent.py`:266--318.)

The loop: two captures and the gap between them — type this (1/2)

```
# Physical (correction-free) slave oscillator: what the hardware
# would do with no synchronization running at all.
physical_slave_frequency = slave.state[1] - slave_frequency_corrections
if settings.sample_clock_offset_ppm is not None:
    sfo_forward = settings.sample_clock_offset_ppm
else:
    sfo_forward = float(
        (physical_slave_frequency - master.state[1]).item()
        / (2.0 * math.pi * settings.carrier_frequency_hz)
        * 1e6
    )
history["physical_relative_frequency"].append(
    (master.state[1] - physical_slave_frequency).clone()
)

relative_state = master.state - slave.state
# Forward frame (master transmits); its intra-frame LO walk becomes
# part of the true state before the reverse frame is exchanged, so
# the two directions see one continuous noise process.
capture_forward = link_forward.capture(master, slave, iteration, sfo_forward)
master.state[0] = wrap_phase(master.state[0] + capture_forward.lo_walk_end)
# Half-duplex turnaround: the radios need real time to switch from
# receive to transmit, and both oscillators keep running across the
# gap - deterministic advance at their current frequencies plus the
# white-FM walk accumulated during it. (The channel is held static
# across the gap; reciprocity aging under mobility remains a known
# simplification.)
turnaround = settings.tdd_turnaround_s
if turnaround > 0.0:
    walk_std = settings.phase_noise_std_rad * math.sqrt(
```

The loop: two captures and the gap between them — type this (2/2)

```
        settings.sample_rate * turnaround
    )
    for oscillator in (master, slave):
        wander = torch.zeros(), dtype=REAL_DTYPE, device=device)
        if walk_std > 0.0:
            wander = (
                torch.randn(
                    (), dtype=REAL_DTYPE, device=device, generator=generator
                )
                * walk_std
            )
            oscillator.state[0] = wrap_phase(
                oscillator.state[0] + oscillator.state[1] * turnaround + wander
            )
        capture_reverse = link_reverse.capture(slave, master, iteration, -sfo_forward)
        slave.state[0] = wrap_phase(slave.state[0] + capture_reverse.lo_walk_end)

    if settings.phase_noise_std_rad > 0.0 and remainder_samples > 0:
        carried_lo_walk = torch.randn(
            (), dtype=REAL_DTYPE, device=device, generator=generator
        ) * (settings.phase_noise_std_rad * math.sqrt(remainder_samples))
    else:
        carried_lo_walk = torch.zeros(), dtype=REAL_DTYPE, device=device)
```

The loop: the half-difference, acquire, track

The central lines of the method. Forward measures $\theta_m - \theta_s + \varphi_{ch}$; reverse measures $\theta_s - \theta_m + \varphi_{ch}$; half the difference cancels the channel exactly (reciprocity). The turnaround drift is backed out with the loop's *own* frequency estimate. Then the filter: predict; tell it about a landed correction; on first detection seed the state from the measurement; afterwards pick the branch against the prediction and update on the circle. No detection $\Rightarrow$ coast. (coherent.py:320--363.)

The loop: the half-difference, acquire, track — type this (1/2)

```
forward = synchronizer.estimate(capture_forward.samples)
reverse = synchronizer.estimate(capture_reverse.samples)
detected = forward.detected and reverse.detected
# Reciprocity: the channel phase is common to both directions and
# cancels in the half-difference, leaving the oscillator offset.
combined_frequency = (forward.frequency - reverse.frequency) / 2.0
# The reverse frame was measured one turnaround later, so the raw
# half-difference contains an extra omega*turnaround/2 of drift. A
# real receiver knows its own turnaround and removes it with its
# measured CFO; the irreducible residue is the CFO-estimate error
# times the gap (plus the random walk across it).
combined_half = wrap_phase(
    wrap_phase(forward.phase - reverse.phase) / 2.0
    + chain_bias
    - combined_frequency * settings.tdd_turnaround_s / 2.0
)

ekf.predict()
if due_correction is not None:
    ekf.reset_after_correction(due_correction)
```

The loop: the half-difference, acquire, track — type this (2/2)

```
if detected:
    if not acquired:
        phase_measurement = _pick_half_phase(
            combined_half, torch.zeros_like(combined_half)
        )
        ekf.state = torch.stack((phase_measurement, combined_frequency))
        ekf.covariance = torch.diag(
            torch.stack((measurement_noise[0, 0], measurement_noise[2, 2]))
        )
        acquired = True
    else:
        phase_measurement = _pick_half_phase(
            combined_half, wrap_phase(ekf.state[0])
        )
        ekf.update(
            torch.stack(
                (
                    torch.cos(phase_measurement),
                    torch.sin(phase_measurement),
                    combined_frequency,
                )
            )
        )
```

The loop: actuate, record, and the figure of merit

The correction that will land L intervals from now should cancel the offset *then* – so the estimate is pushed L steps through the transition matrix, quantized to what hardware can apply, and scheduled; the filter reset happens when it *lands*. The scoreboard is the true residual between the clocks – something only a simulation can see – and the coherent gain it implies: $|1 + e^{j\text{residual}}|^2/4 = \cos^2(\text{residual}/2)$. (`coherent.py:364--420`.)

The loop: actuate, record, and the figure of merit — type this (1/2)

```
    predicted = ekf.state.clone()
    for _ in range(settings.correction_latency_intervals):
        predicted = ekf.transition @ predicted
        correction = _quantize_correction(predicted, settings)
    else:
        phase_measurement = combined_half
        correction = None

    true_phase = wrap_phase(relative_state[0])
    estimated_phase = wrap_phase(ekf.state[0])
    history["true_phase"].append(true_phase.clone())
    history["measured_phase"].append(phase_measurement.clone())
    history["estimated_phase"].append(estimated_phase.clone())
    history["true_frequency"].append(relative_state[1].clone())
    history["estimated_frequency"].append(ekf.state[1].clone())
    history["phase_error"].append(wrap_phase(true_phase - estimated_phase))
    history["frequency_error"].append(
        (relative_state[1] - ekf.state[1]).clone()
    )
    history["detected"].append(
        torch.tensor(detected, dtype=torch.bool, device=device)
    )
```


The loop: actuate, record, and the figure of merit — type this (2/2)

```
if correction is not None:
    if settings.correction_latency_intervals == 0:
        slave.apply_correction(correction)
        ekf.reset_after_correction(correction)
        slave_frequency_corrections = (
            slave_frequency_corrections + correction[1]
        )
        correction_has_loaded = True
    else:
        pending_corrections[
            iteration + settings.correction_latency_intervals
        ] = correction

history["correction_active"].append(
    torch.tensor(correction_has_loaded, dtype=torch.bool, device=device)
)
history["calibrated"].append(
    torch.tensor(pi_calibrated, dtype=torch.bool, device=device)
)
residual_state = master.state - slave.state
residual_phase = wrap_phase(residual_state[0])
history["post_correction_phase"].append(residual_phase.clone())
history["post_correction_frequency"].append(residual_state[1].clone())
# Two-element open-loop combining toward a passive target:
#  $|1 + \exp(j * \text{residual})|^2 / 4 = \cos^2(\text{residual} / 2)$ .
history["coherent_gain"].append(torch.cos(residual_phase / 2.0).square())

return TwoWaySimulationResult(
    **{
        name: torch.stack(values).detach().cpu() for name, values in history.items()
    },
    airtime_fraction=airtime_fraction,
    device=device,
)
```

Run it

The finished program. Two clocks starting 1500 Hz and 1.2 rad apart, a 3GPP multipath channel, the full hardware impairment chain – and the loop holds them aligned to a few degrees. Because every line above is the repository's own code, these numbers are identical to `ota_sync.run_two_way_simulation`'s output, bit for bit.

Run it — type this

```
if __name__ == "__main__":  
    result = run_two_way_simulation(SDRSimulationConfig())  
    print(f"detection rate      : {result.detection_rate:.2f}")  
    print(f"steady phase rms    : {result.steady_state_phase_rms:.4f} rad")  
    print(f"mean coherent gain    : {result.mean_coherent_gain * 100:.1f} %")  
    print(f"airtime fraction        : {result.airtime_fraction * 100:.2f} %")  
    print(f"final freq error         : {result.final_frequency_error_hz:.3f} Hz")
```

Verify: your file IS the repository's code

Your build reproduces the repository's simulation **bit for bit** — same random draws, same channel, same every-interval state. The check (run from `coding_guide/`):

```
import sys; sys.path.insert(0, "..")
import torch, build_twoway as bt
from ota_sync import (run_two_way_simulation,
                      SDRSimulationConfig)
a = bt.run_two_way_simulation(bt.SDRSimulationConfig())
b = run_two_way_simulation(SDRSimulationConfig())
print(torch.equal(a.post_correction_phase,
                  b.post_correction_phase),
      torch.equal(a.coherent_gain, b.coherent_gain))
```

Output when this guide was generated: True True.

And the numbers your file prints (identical to the package):

```
detection rate      : 1.00
steady phase rms    : 0.0835 rad
mean coherent gain  : 99.8 %
airtime fraction    : 19.12 %
final freq error     : -0.010 Hz
```

What you built, in one pass

Per 50 ms interval, your code now does — because you typed every line of it:

- 1 Both clocks random-walk; the master absorbs the carried intra-frame walk and the flicker step.
- 2 A latency-delayed correction lands, if due; the filter is told.
- 3 (Once, after three settled corrections) the π power check repairs a wrong acquisition branch.
- 4 Forward capture through the full impairment chain; 1 ms turnaround; reverse capture.
- 5 Both captures estimated; the half-difference cancels the channel by reciprocity; turnaround drift backed out.
- 6 Filter: predict $\rightarrow$ branch pick $\rightarrow$ circle-measurement update — or coast on a failed detection.
- 7 The correction is predicted forward by the command latency, quantized, and scheduled.
- 8 Truth is recorded: residual phase and $\cos^2(\text{residual}/2)$.

Part II — the action items from the last meeting

Part II is the agreed work for the next meeting, item by item. A second file, `build_sensing.py`, same rules as Part I: the snippets, typed in order, are the complete file; every number on these slides is its measured output.

- ❶ *“Implementing timing for LTF/STF for frequency/phase estimation ...building it from the ground up to get a better intuition”* $\Rightarrow$ the short training field (STF) and long training field (LTF) sections: preamble construction and the three-stage receiver, written bare-handed and tested over 500 impaired trials.
- ❷ *“Implementing realistic parameters (carrier f , BW, etc.) to model real base stations”* $\Rightarrow$ the `BaseStationParams` section: 3.5 GHz carrier, 20 MHz bandwidth, 10 W, 8 dBi, 7 dB noise figure, and the -94 dBm floor they imply.
- ❸ *“For a single base station, monostatic sensing in LOS and NLOS cases in Sionna”* $\Rightarrow$ the three ray-traced scenes: line-of-sight, and two NLOS cases — fully blocked, and blocked with a reflected detour (the ghost target).
- ❹ *“Compare OFDM waveforms to single tone waveforms and assess their performance”* $\Rightarrow$ the waveform section: counted frequency and delay accuracy for both.

And the meeting's third pillar — all of this *for the purpose of passive target detection and localization* — closes the part: four stations localize a drone, measured against the Cramér–Rao bound.

Part II — practical note

The ray-traced sections need Sionna's ray-tracing module. On this machine that lives in the project environment — run Part II with

```
~/Downloads/Princeton_Research/ota_sync/ .venv/bin/python  
build_sensing.py
```

The training-field and waveform sections run under any Python with numpy.

Part II file header

Create `build_sensing.py`. Plain numpy plus two physical constants; Sionna's ray tracer is imported later, inside the function that needs it, so the non-ray-traced sections run anywhere.

Part II file header — type this

```
"""Part II of the coding guide: realistic base-station parameters,  
short/long training fields built from the ground up, single-station  
monostatic sensing in Sionna (line-of-sight and blocked), and OFDM  
versus single-tone waveforms.
```

```
New code, written for this guide and tested; every measured number  
quoted on the slides comes from running this file:
```

```
python3 build_sensing.py  
"""  
  
from __future__ import annotations  
  
import math  
import os  
import tempfile  
from dataclasses import dataclass  
  
import numpy as np  
  
BOLTZMANN_TO = 1.380649e-23 * 290.0  
SPEED_OF_LIGHT = 299792458.0  
  
rng = np.random.default_rng(0)
```

Realistic base-station parameters

One dataclass models a mid-band 5G small cell: 3.5 GHz carrier (band n78), 20 MHz of bandwidth sampled at the bandwidth (complex baseband), 10 W transmit power, an 8 dBi antenna, a 7 dB receiver noise figure. The one derived number that drives everything: the thermal noise power $kT_0 \cdot F \cdot B$. At 20 MHz that is -94 dBm – every detection question below is a contest between an echo and this floor.

Realistic base-station parameters — type this

```
@dataclass(frozen=True)
class BaseStationParams:
    carrier_frequency_hz: float = 3.5e9      # 5G band n78
    bandwidth_hz: float = 20e6               # sample rate = bandwidth
    tx_power_w: float = 10.0                 # small-cell class
    antenna_gain_dbi: float = 8.0
    noise_figure_db: float = 7.0
    subcarriers: int = 64                    # OFDM grid over the band
    cyclic_prefix: int = 16

    @property
    def wavelength_m(self) -> float:
        return SPEED_OF_LIGHT / self.carrier_frequency_hz

    @property
    def sample_period_s(self) -> float:
        return 1.0 / self.bandwidth_hz

    @property
    def noise_power_w(self) -> float:
        return (BOLTZMANN_TO
            * 10.0 ** (self.noise_figure_db / 10.0)
            * self.bandwidth_hz)
```

The short and long training fields, from the ground up

The same two-field idea as the sync preamble, now built bare-handed at base-station parameters (the 802.11/LTE pattern). **Short field:** one 16-sample constant-power sequence tiled ten times – its job is coarse timing and coarse frequency. **Long field:** a known 64-subcarrier BPSK symbol, sent twice behind one 32-sample cyclic prefix – its job is fine frequency and the phase. At 20 MHz the 16-sample repetition makes coarse frequency unambiguous to ± 625 kHz; a ± 10 ppm oscillator at 3.5 GHz is only ± 35 kHz, so acquisition always succeeds.

The short and long training fields, from the ground up — type this

```
STF_LEN = 16
STF_REPS = 10
LTF_LEN = 64
LTF_CP = 32

def make_training_fields(params: BaseStationParams, seed: int = 7):
    """Return (waveform, ltf_time) -- the transmitted preamble and
    the known long-field symbol the receiver matches against."""
    field_rng = np.random.default_rng(seed)
    # Short field: constant-modulus random-QPSK 16-sample sequence.
    stf = np.exp(1j * (np.pi / 2.0)
                 * field_rng.integers(0, 4, STF_LEN)
                 + 1j * np.pi / 4.0)
    # Long field: known BPSK on all 64 subcarriers, to time domain.
    bpsk = 2.0 * field_rng.integers(0, 2, LTF_LEN) - 1.0
    ltf_time = np.fft.ifft(bpsk) * math.sqrt(LTF_LEN)
    waveform = np.concatenate([
        np.tile(stf, STF_REPS),
        ltf_time[-LTF_CP:],          # cyclic prefix
        ltf_time,
        ltf_time,                    # the repeat that measures CFO
    ])
    return waveform, ltf_time
```

The receiver: timing, frequency, phase

Three stages, each earning one quantity. (1) **Coarse (Schmidl–Cox)**: correlate the signal with itself 16 samples later; the correlation magnitude peaks where the repeating short field coherently fills the window (coarse timing) and its angle is the frequency offset accumulated over 16 samples. (Deliberately the *unnormalized* peak: the classic normalized ratio spikes on noise-only windows – found the hard way while testing this.) (2) **Fine timing**: after coarse derotation, cross-correlate with the known long symbol – a single sharp peak. (3) **Fine frequency and phase**: the two long symbols are identical and 64 samples apart; their angle refines the frequency, and the matched filter of both, after derotation, reads the carrier phase.

The receiver: timing, frequency, phase — type this (1/2)

```
def estimate_preamble(rx, ltf_time, params: BaseStationParams):
    """Return (timing_index, cfo_hz, phase_rad)."""
    ts = params.sample_period_s
    stf_span = STF_LEN * STF_REPS

    # (1) coarse: lag-16 self-correlation over the short field.
    lag = STF_LEN
    width = stf_span - lag
    prod = np.conj(rx[:-lag]) * rx[lag:]
    csum = np.cumsum(np.concatenate([[0.0 + 0j], prod]))
    corr = csum[width:] - csum[:-width]
    # Unnormalized peak: the ratio metric spikes on noise-only
    # windows (tiny numerator over tiny denominator); the raw
    # correlation magnitude only peaks where the repeats coherently
    # fill the window.
    coarse = int(np.argmax(np.abs(corr)))
    cfo = np.angle(corr[coarse]) / (lag * ts) / (2.0 * np.pi) # Hz

    # (2) fine timing: matched filter of the known long symbol.
    # The long field repeats, so the filter has TWO equal peaks, 64
    # samples apart -- score candidate positions by the pair
    # |mf[i]| + |mf[i+64]|, which only the FIRST symbol maximizes.
    n = np.arange(rx.size)
    derot = rx * np.exp(-2j * np.pi * cfo * n * ts)
    mf = np.abs(np.correlate(derot, ltf_time, mode="valid"))
    # Search a window around where the long field should start,
    # with a margin on BOTH sides -- the coarse estimate can land a
    # sample or two late, which would otherwise push the true peak
    # just outside the window.
    start = max(coarse + stf_span + LTF_CP - 8, 0)
```

The receiver: timing, frequency, phase — type this (2/2)

```
window = mf[start : min(start + 4 * LTF_LEN, mf.size)]
pair = window[:-LTF_LEN] + window[LTF_LEN:]
first_sym = start + int(np.argmax(pair))
first_sym = min(first_sym, rx.size - 2 * LTF_LEN)

# (3) fine frequency from the symbol repeat, then the phase.
a = derot[first_sym : first_sym + LTF_LEN]
b = derot[first_sym + LTF_LEN : first_sym + 2 * LTF_LEN]
fine = np.angle(np.sum(np.conj(a) * b)) / (LTF_LEN * ts) \
    / (2.0 * np.pi)
cfo += fine
derot = rx * np.exp(-2j * np.pi * cfo * n * ts)
both = derot[first_sym : first_sym + 2 * LTF_LEN]
phase = np.angle(np.sum(np.conj(np.tile(ltf_time, 2)) * both))
return first_sym, cfo, phase
```


Test the receiver at realistic impairments

The test: random arrival time, a 20 kHz frequency offset (≈ 6 ppm at 3.5 GHz), a fixed 3-tap multipath channel, 15 dB signal-to-noise ratio, 500 trials. Measured: timing exact in **500/500** trials; frequency error **1.03 kHz rms** (0.29 ppm of the carrier); phase repeatable to **0.107 rad rms**. The kHz-level frequency error is not a flaw – it is physics: frequency precision is bought with observation *time*, and this whole preamble lasts 16 μ s. Part I's synchronization preamble reached ~ 0.1 Hz because it observes for 4 ms, at $250\times$ less bandwidth. Time buys frequency; bandwidth buys delay – hold that thought for the waveform comparison.

Test the receiver at realistic impairments — type this

```
def test_training_fields(params: BaseStationParams, trials=500,
                        snr_db=15.0, cfo_hz=20e3):
    waveform, ltf_time = make_training_fields(params)
    ts = params.sample_period_s
    # One fixed multipath channel across trials, so the phase spread
    # measures the ESTIMATOR, not a re-rolled channel.
    taps = np.array([1.0,
                     0.4 * np.exp(2j * np.pi * 0.3),
                     0.2 * np.exp(2j * np.pi * 0.7)])
    timing_hits, cfo_errs, phases = 0, [], []
    for _ in range(trials):
        delay = int(rng.integers(20, 120))
        tx = np.zeros(delay + waveform.size + 96, complex)
        tx[delay : delay + waveform.size] = waveform
        rx = (np.convolve(tx, taps)[: tx.size]
              * np.exp(2j * np.pi * cfo_hz
                       * np.arange(tx.size) * ts))
        snr = 10.0 ** (snr_db / 10.0)
        sigma = math.sqrt(np.mean(np.abs(waveform) ** 2)
                          / (2.0 * snr))
        rx = rx + sigma * (rng.standard_normal(rx.size)
                           + 1j * rng.standard_normal(rx.size))
        sym, cfo, phase = estimate_preamble(rx, ltf_time, params)
        expected = delay + STF_LEN * STF_REPS + LTF_CP
        timing_hits += int(sym == expected)
        cfo_errs.append(cfo - cfo_hz)
        phases.append(phase)
    # Circular spread: length of the mean phasor.
    resultant = np.abs(np.mean(np.exp(1j * np.array(phases))))
    phase_std = math.sqrt(max(-2.0 * math.log(resultant), 0.0))
    print(f" timing exact      : {timing_hits}/{trials}")
    print(f" cfo error rms      : {np.std(cfo_errs):.1f} Hz ")
    print(f"      ({np.std(cfo_errs)/params.carrier_frequency_hz*1e6:.3f} ppm)")
    print(f" phase spread rms       : {phase_std:.4f} rad")
```

The ray-tracing model

What Sionna's ray tracer actually computes, because Part II leans on it. A **scene** is a set of triangle meshes, each wearing a **radio material**: a complex permittivity (its real part ε_r and conductivity σ_c set, at the carrier frequency, how much of an incident wave reflects and with what phase – the Fresnel coefficients) and a thickness. The **path solver** then finds every geometric route from a transmitter to a receiver up to a bounce budget (`max_depth`): the direct line if unobstructed, and specular (mirror-like) reflections found by the image method – reflect the endpoint across a face, draw the straight line, keep it if nothing blocks it. Every found path is returned as one complex gain a_p (spreading loss $\times$ the Fresnel losses of its bounces, phase = electrical length) and one delay τ_p – exactly the (a_p, τ_p) pairs the echo model consumes. Occlusion is decided by the actual triangles: put a building in the way and the direct path simply does not appear in the list. The two helpers below write scene geometry (a ground plane, a box building) as PLY mesh files the scene loader accepts.

The ray-tracing model — type this

```
def _ply_box(x, y, w, d, h):
    x0, x1, y0, y1 = x - w / 2, x + w / 2, y - d / 2, y + d / 2
    v = [(x0, y0, 0), (x1, y0, 0), (x1, y1, 0), (x0, y1, 0),
          (x0, y0, h), (x1, y0, h), (x1, y1, h), (x0, y1, h)]
    f = [(0, 1, 5), (0, 5, 4), (1, 2, 6), (1, 6, 5), (2, 3, 7),
          (2, 7, 6), (3, 0, 4), (3, 4, 7), (4, 5, 6), (4, 6, 7)]
    lines = ["ply", "format ascii 1.0", f"element vertex {len(v)}",
             "property float x", "property float y",
             "property float z", f"element face {len(f)}",
             "property list uchar int vertex_indices", "end_header"]
    lines += [f"{a} {b} {c}" for a, b, c in v]
    lines += [f"3 {a} {b} {c}" for a, b, c in f]
    return "\n".join(lines) + "\n"

def _ply_ground(half):
    return ("ply\nformat ascii 1.0\nelement vertex 4\n"
            "property float x\nproperty float y\nproperty float z\n"
            "element face 2\n"
            "property list uchar int vertex_indices\nend_header\n"
            f"-{half} -{half} 0\n{half} -{half} 0\n"
            f"{half} {half} 0\n-{half} {half} 0\n"
            "3 0 1 2\n3 0 2 3\n")
```

Monostatic sensing: three scenes

One base station (mast at 15 m), one passive target (40 m altitude, 300 m down-range), no cooperation: the station transmits and listens for its own echo. The ray tracer supplies the station $\rightarrow$ target leg with a point probe receiver at the target position (a ray tracer cannot hit a target-sized object at range, so the standard coupling applies the radar cross-section analytically and reuses the leg for the round trip). Three scenes on the same geometry: **line-of-sight** (ground plane only); **blocked** (a $40 \times 40 \times 60$ m concrete building midway); **blocked + reflector** (same blocker, plus a long building row 80 m to the side whose wall offers the solver a specular detour around the blocker – the urban canyon situation).

Monostatic sensing: three scenes — type this (1/3)

```
def monostatic_scene(params: BaseStationParams, scenario: str):
    """Build one of the three scenes; returns (rt, scene)."""
    import sionna.rt as rt

    scene = rt.load_scene()
    scene.frequency = params.carrier_frequency_hz
    scene.tx_array = rt.PlanarArray(num_rows=1, num_cols=1,
                                    pattern="iso", polarization="V")
    scene.rx_array = scene.tx_array

    def add(name, ply, material):
        handle = tempfile.NamedTemporaryFile("w", suffix=".ply",
                                             delete=False)
        handle.write(ply)
        handle.close()
        try:
            scene.edit(add=rt.SceneObject(
                fname=handle.name, name=name, radio_material=material))
        finally:
            os.unlink(handle.name)
```

Monostatic sensing: three scenes — type this (2/3)

```
ground = rt.RadioMaterial("ground", thickness=10.0,
                          relative_permittivity=15.0,
                          conductivity=0.035)
add("ground-plane", _ply_ground(800.0), ground)
concrete = rt.RadioMaterial("concrete", thickness=0.3,
                           relative_permittivity=5.24,
                           conductivity=0.123)
if scenario in ("blocked", "reflector"):
    add("blocker", _ply_box(150.0, 0.0, 40.0, 40.0, 60.0),
        concrete)
if scenario == "reflector":
    # A long building row whose street-side wall (y = 80 m) can
    # mirror the signal around the blocker.
    add("reflector", _ply_box(150.0, 90.0, 200.0, 20.0, 60.0),
        concrete)

scene.add(rt.Transmitter("bs", position=[0.0, 0.0, 15.0]))
scene.add(rt.Receiver("target-probe",
                     position=[300.0, 0.0, 40.0]))
return rt, scene
```

Monostatic sensing: three scenes — type this (3/3)

```
def monostatic_legs(rt, scene):  
    """Solve the scene: returns (paths, gains, delays)."""  
    paths = rt.PathSolver()(scene, max_depth=3, los=True,  
                             specular_reflection=True,  
                             diffuse_reflection=False,  
                             refraction=False)  
    a, tau = paths.cir(normalize_delays=False, out_type="numpy")  
    if a.shape[-2] == 0:  
        # The solver found no way from station to target at all --  
        # complete blockage. Zero paths means zero echo.  
        return paths, np.zeros(0, complex), np.zeros(0)  
    # a: [rx, rx_ant, tx, tx_ant, path, time]; tau may or may not  
    # keep the antenna axes depending on the array setup.  
    return (paths, a[0, 0, 0, 0, :, 0],  
            tau.reshape(-1, tau.shape[-1])[0])
```


Monostatic sensing: echo, matched filter, range

The echo is assembled from the traced paths: going out on path p and back on path q contributes amplitude $a_p a_q \sqrt{4\pi\sigma}/\lambda$ at delay $\tau_p + \tau_q$ (radar cross-section $\sigma = 1 \text{ m}^2$, a small vehicle). Fractional delays are applied exactly, as a phase ramp in the frequency domain. Thermal noise at the -94 dBm floor, matched filter against the transmitted burst, parabolic interpolation around the peak, and range $= c\hat{\tau}/2$. Note what detection will hinge on: the echo arrives *below* the noise floor per sample; only the matched filter's processing gain ($10 \log_{10} 1600 = 32 \text{ dB}$ over the burst) lifts it out.

Monostatic sensing: echo, matched filter, range — type this (1/2)

```
def fractional_delay(signal, delay_samples, total_len):
    padded = np.zeros(total_len, complex)
    padded[: signal.size] = signal
    freq = np.fft.fftfreq(total_len)
    return np.fft.ifft(np.fft.fft(padded)
                       * np.exp(-2j * np.pi * freq * delay_samples))

def monostatic_range(params: BaseStationParams, burst,
                    gains, delays, trials=100):
    """Counted range estimates; returns (detect_rate, errors_m,
    echo power in watts)."""
    ts = params.sample_period_s
    amp_gain = 10.0 ** (params.antenna_gain_dbi / 20.0)
    rcs_factor = math.sqrt(4.0 * math.pi * 1.0) \
        / params.wavelength_m
    total = burst.size + 4096
    echo = np.zeros(total, complex)
    for p in range(gains.size):
        for q in range(gains.size):
            amplitude = (math.sqrt(params.tx_power_w) * amp_gain**2
                        * gains[p] * gains[q] * rcs_factor)
            delay_s = (delays[p] + delays[q]) / ts
            echo += amplitude * fractional_delay(
                burst, delay_s, total)
    echo_power = float(np.mean(np.abs(echo) ** 2)
                      * total / burst.size)
    sigma = math.sqrt(params.noise_power_w / 2.0)
    true_range = math.sqrt(300.0**2 + 25.0**2)
```

Monostatic sensing: echo, matched filter, range — type this (2/2)

```
detections, errors = 0, []
for _ in range(trials):
    rx = echo + sigma * (rng.standard_normal(total)
                        + 1j * rng.standard_normal(total))
    mf = np.abs(np.correlate(rx, burst, mode="valid")) ** 2
    peak = int(np.argmax(mf))
    floor = np.median(mf)
    if mf[peak] < 30.0 * floor:      # -15 dB above the floor
        continue
    detections += 1
    if 0 < peak < mf.size - 1:      # parabolic refinement
        num = mf[peak - 1] - mf[peak + 1]
        den = mf[peak - 1] - 2 * mf[peak] + mf[peak + 1]
        peak = peak + 0.5 * num / den
    range_m = peak * ts * SPEED_OF_LIGHT / 2.0
    errors.append(range_m - true_range)
return detections / trials, np.array(errors), echo_power
```

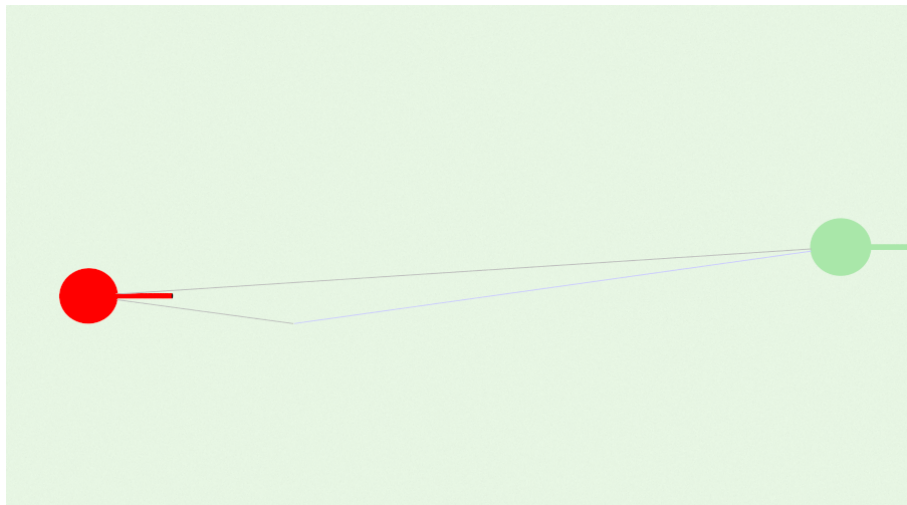
Render the scenes, with the traced rays

The ray tracer can draw exactly what it solved: `render_to_file` paints the scene geometry from a camera and overlays the found propagation paths. The three renders on the next slides are the three scenarios, with every path the solver found drawn in. Line-of-sight: the direct ray and the ground bounce. Blocked: the building swallows everything – no rays to draw. Blocked + reflector: the detour around the blocker via the building row's wall, plus its ground-bounce variants.

Render the scenes, with the traced rays — type this

```
def render_scenes(params: BaseStationParams, out_dir="figures"):
    os.makedirs(out_dir, exist_ok=True)
    for scenario in ("los", "blocked", "reflector"):
        rt, scene = monostatic_scene(params, scenario)
        paths, gains, _ = monostatic_legs(rt, scene)
        camera = rt.Camera(position=[150.0, -340.0, 300.0],
                             look_at=[150.0, 20.0, 20.0])
        scene.render_to_file(
            camera=camera,
            filename=os.path.join(out_dir,
                                   f"scene_{scenario}.png"),
            # The renderer cannot overlay an empty path list.
            paths=paths if gains.size else None,
            resolution=(900, 500))
```

Render the scenes, with the traced rays



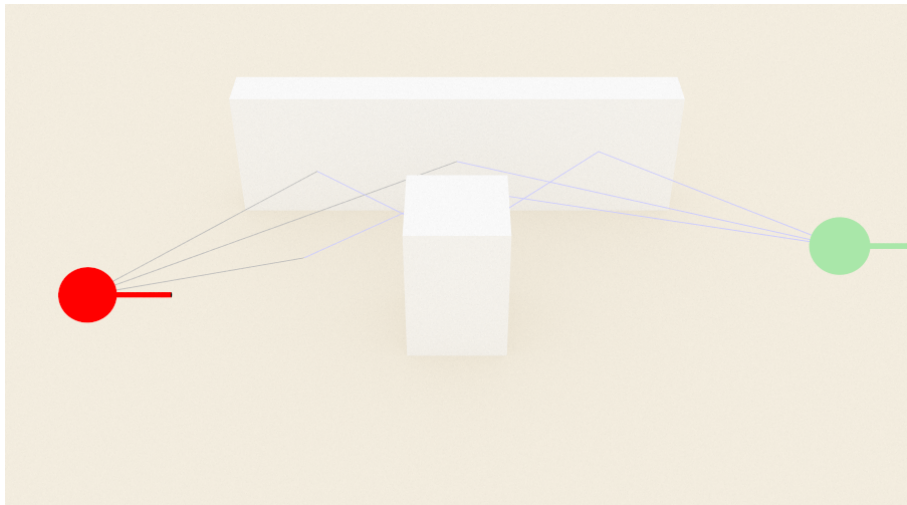
Line-of-sight: the direct ray and the ground bounce reach the target.

Render the scenes, with the traced rays



Blocked: the solver finds no path at all – there are no rays to draw.

Render the scenes, with the traced rays



Blocked + reflector: the wall of the building row mirrors the signal around the blocker (three paths, all longer than the straight line).

The range profiles, side by side

One noisy trial per scenario, matched filter output against range. Line-of-sight: one clean peak at the true 301 m. Blocked: noise, nothing to find. Blocked + reflector: a real peak – but at the detour's length, not the target's range. The echo survives the blockage and the detection is genuine; the *range* is a lie (about +40 m here). A single station cannot tell a detour from a distance; resolving that ambiguity takes more geometry – stations that look from somewhere else.

The range profiles, side by side — type this (1/2)

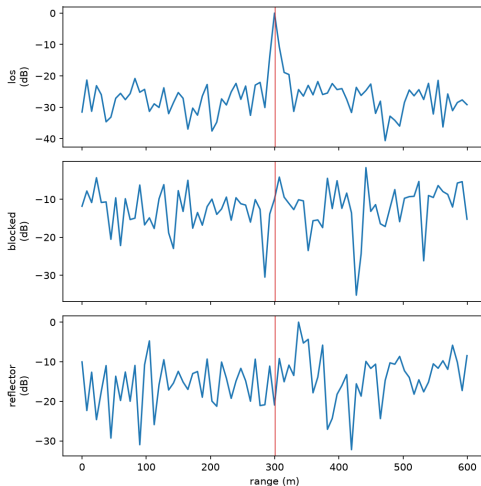
```
def range_profile_figure(params: BaseStationParams,
                        out_dir="figures"):
    import matplotlib
    matplotlib.use("Agg")
    import matplotlib.pyplot as plt

    os.makedirs(out_dir, exist_ok=True)
    ts = params.sample_period_s
    burst = make_ofdm_burst(params)
    sigma = math.sqrt(params.noise_power_w / 2.0)
    true_range = math.sqrt(300.0**2 + 25.0**2)
    fig, axes = plt.subplots(3, 1, figsize=(7, 7), sharex=True)
    for axis, scenario in zip(axes,
                              ("los", "blocked", "reflector")):
        rt, scene = monostatic_scene(params, scenario)
        _, gains, delays = monostatic_legs(rt, scene)
        total = burst.size + 4096
        echo = np.zeros(total, complex)
        amp_gain = 10.0 * (params.antenna_gain_dbi / 20.0)
        rcs_factor = math.sqrt(4.0 * math.pi) / params.wavelength_m
        for p in range(gains.size):
            for q in range(gains.size):
                amplitude = (math.sqrt(params.tx_power_w)
                             * amp_gain**2 * gains[p] * gains[q]
                             * rcs_factor)
                echo += amplitude * fractional_delay(
                    burst, (delays[p] + delays[q]) / ts, total)
    rx = echo + sigma * (rng.standard_normal(total)
                        + 1j * rng.standard_normal(total))
    mf = np.abs(np.correlate(rx, burst, mode="valid")) ** 2
```

The range profiles, side by side — type this (2/2)

```
ranges = (np.arange(mf.size) * ts * SPEED_OF_LIGHT / 2.0)
keep = ranges <= 600.0
axis.plot(ranges[keep],
          10.0 * np.log10(mf[keep] / np.max(mf)))
axis.axvline(true_range, color="C3", linewidth=0.8)
axis.set_ylabel(f"{scenario}\n(dB)")
axes[-1].set_xlabel("range (m)")
fig.tight_layout()
fig.savefig(os.path.join(out_dir, "range_profiles.png"),
            dpi=150)
```

The range profiles, side by side



Matched-filter range profiles, one noisy trial per scenario; the vertical line is the true range.

OFDM versus a single tone

Same duration, same power, two waveforms. The **OFDM burst** spreads energy over the whole 20 MHz; the **single tone** concentrates it at one frequency. For *frequency and phase* the tone is fine – fit a phase ramp. For *range* the tone is nearly blind: range lives in *delay*, delay precision comes from *bandwidth* ($c/2B = 7.5$ m at 20 MHz), and the carrier itself carries none. What little timing a real tone burst has lives in its slow power envelope – which is why the tone here gets a smooth ramp: a rectangular edge would smuggle wideband timing into a “narrowband” waveform. This is the cleanest statement of why sensing wants wideband waveforms while synchronization alone can live on a tone.

OFDM versus a single tone — type this (1/2)

```
def make_ofdm_burst(params: BaseStationParams, symbols=20, seed=3):
    burst_rng = np.random.default_rng(seed)
    angles = (burst_rng.integers(0, 4,
                                (symbols, params.subcarriers))
              * (np.pi / 2.0) + np.pi / 4.0)
    time_symbols = np.fft.iffreq(np.exp(1j * angles), axis=1) \
        * math.sqrt(params.subcarriers)
    with_cp = np.concatenate(
        [time_symbols[:, -params.cyclic_prefix:], time_symbols],
        axis=1).reshape(-1)
    return with_cp / math.sqrt(np.mean(np.abs(with_cp) ** 2))

def compare_waveforms(params: BaseStationParams, trials=300,
                      snr_db=10.0, cfo_hz=5e3):
    ts = params.sample_period_s
    ofdm = make_ofdm_burst(params)
    # A single tone at one subcarrier frequency. A real transmitter
    # ramps power smoothly, so the tone gets a smooth (Hann)
    # envelope -- a rectangular edge would smuggle wideband timing
    # information into a "narrowband" waveform.
    tone = (np.exp(2j * np.pi * 3.125e6
                  * np.arange(ofdm.size) * ts)
            * np.hanning(ofdm.size))
    tone = tone / math.sqrt(np.mean(np.abs(tone) ** 2))
    snr = 10.0 ** (snr_db / 10.0)
    sigma = math.sqrt(1.0 / (2.0 * snr))
    results = {}
    for name, burst in (("ofdm", ofdm), ("tone", tone)):
        cfo_errs, delay_errs = [], []
```

OFDM versus a single tone — type this (2/2)

```
for _ in range(trials):
    delay = int(rng.integers(0, 200))
    rx = fractional_delay(burst, delay, burst.size + 512)
    rx = rx * np.exp(2j * np.pi * cfo_hz
                    * np.arange(rx.size) * ts)
    rx = rx + sigma * (rng.standard_normal(rx.size)
                      + 1j * rng.standard_normal(rx.size))
    # frequency: split-correlation, identical for both
    half = burst.size // 2
    seg = rx[delay : delay + burst.size]
    r = np.sum(np.conj(burst[:half] * seg[half:half+half])
              * (burst[half:half+half] * np.conj(seg[half:half])))
    est = -np.angle(r) / (half * ts) / (2.0 * np.pi)
    cfo_errs.append(est - cfo_hz)
    # delay: matched filter, no help given
    mf = np.abs(np.correlate(rx, burst, mode="valid"))
    delay_errs.append(int(np.argmax(mf)) - delay)
    results[name] = (np.std(cfo_errs),
                    np.sqrt(np.mean(np.square(delay_errs))))
for name, (cfo_rms, delay_rms) in results.items():
    print(f" {name:5s} cfo rms {cfo_rms:8.1f} Hz | "
          f"delay rms {delay_rms:8.1f} samples "
          f"({delay_rms * ts * SPEED_OF_LIGHT / 2.0:.0f} m)")
```

Localization: from ranges to a position

The ghost-range result begs the next step: more stations. Four base stations at the corners of the area each run the monostatic measurement on the same **drone** (radar cross-section 0.03 m^2 , hovering at 60 m). Each station's round-trip delay gives one range – a sphere the drone must sit on – and four spheres pin the position. One ray-tracer solve supplies all four legs (ground bounce included); the position solver is Gauss–Newton: linearize each range around the current guess (the gradient of $\|p - s_i\|$ is the unit vector from station to drone), solve the little least-squares problem, step, repeat.

Localization: from ranges to a position — type this (1/4)

```
DRONE = np.array([170.0, 90.0, 60.0])
DRONE_RCS_M2 = 0.03
STATIONS = np.array([[0.0, 0.0, 15.0], [300.0, 0.0, 15.0],
                    [0.0, 200.0, 15.0], [300.0, 200.0, 15.0]])

def localization_legs(params: BaseStationParams):
    """One RT solve, four stations -> drone: [(gains, delays)]."""
    import sionna.rt as rt

    scene = rt.load_scene()
    scene.frequency = params.carrier_frequency_hz
    scene.tx_array = rt.PlanarArray(num_rows=1, num_cols=1,
                                    pattern="iso", polarization="V")
    scene.rx_array = scene.tx_array
    handle = tempfile.NamedTemporaryFile("w", suffix=".ply",
                                         delete=False)

    handle.write(_ply_ground(800.0))
    handle.close()

    try:
        scene.edit(add=rt.SceneObject(
            fname=handle.name, name="ground-plane",
            radio_material=rt.RadioMaterial(
                "ground", thickness=10.0,
                relative_permittivity=15.0, conductivity=0.035)))

    finally:
        os.unlink(handle.name)

    for index, station in enumerate(STATIONS):
        scene.add(rt.Transmitter(f"bs-{index}",
                                position=station.tolist()))
```

Localization: from ranges to a position — type this (2/4)

```
scene.add(rt.Receiver("drone", position=DRONE.tolist()))
paths = rt.PathSolver()(scene, max_depth=3, los=True,
                        specular_reflection=True,
                        diffuse_reflection=False,
                        refraction=False)
a, tau = paths.cir(normalize_delays=False, out_type="numpy")
legs = []
for index in range(len(STATIONS)):
    gains = a[0, 0, index, 0, :, 0]
    delays = tau[0, index, :]
    keep = delays >= 0.0          # padded slots carry tau = -1
    legs.append((gains[keep], delays[keep]))
return legs
```

```
def measure_range(params: BaseStationParams, burst,
                 gains, delays, upsample=16):
    """One noisy monostatic range measurement from one station.
```

The peak search runs on a 16x frequency-domain upsampled cross-correlation: the coarse three-point parabola on the raw sample grid leaves a bias worth several times the noise floor, and an estimator meant to be compared against the Cramer-Rao bound has to earn it.

```
    """
    ts = params.sample_period_s
    amp_gain = 10.0 ** (params.antenna_gain_dbi / 20.0)
    rcs_factor = (math.sqrt(4.0 * math.pi * DRONE_RCS_M2)
                  / params.wavelength_m)
    total = burst.size + 2048
```

Localization: from ranges to a position — type this (3/4)

```
echo = np.zeros(total, complex)
for p in range(gains.size):
    for q in range(gains.size):
        amplitude = (math.sqrt(params.tx_power_w) * amp_gain**2
                     * gains[p] * gains[q] * rcs_factor)
        echo += amplitude * fractional_delay(
            burst, (delays[p] + delays[q]) / ts, total)
sigma = math.sqrt(params.noise_power_w / 2.0)
rx = echo + sigma * (rng.standard_normal(total)
                    + 1j * rng.standard_normal(total))
spectrum = np.fft.fft(rx) * np.conj(np.fft.fft(burst, total))
padded = np.zeros(total * upsample, complex)
padded[: total // 2] = spectrum[: total // 2]
padded[-total // 2 :] = spectrum[-total // 2 :]
cc = np.abs(np.fft.ifft(padded)) ** 2
search = cc[: (total - burst.size) * upsample]
peak = int(np.argmax(search))
num = cc[peak - 1] - cc[peak + 1]
den = cc[peak - 1] - 2 * cc[peak] + cc[peak + 1]
peak = peak + 0.5 * num / den
return peak / upsample * ts * SPEED_OF_LIGHT / 2.0, echo
```

```
def solve_position(ranges, stations, initial):
    """Gauss-Newton: least-squares position from station ranges.
```

Localization: from ranges to a position — type this (4/4)

The initial guess must sit OFF the stations' common plane:
every look direction's vertical component is zero there, so the
plane is a stationary point the iteration can never leave --
found the hard way (the solver returned $z = 15$ m forever).

```
"""
position = initial.copy()
for _ in range(20):
    vectors = position - stations
    distances = np.linalg.norm(vectors, axis=1)
    jacobian = vectors / distances[:, None]
    residual = ranges - distances
    step, *_ = np.linalg.lstsq(jacobian, residual, rcond=None)
    position = position + step
    if np.linalg.norm(step) < 1e-6:
        break
return position
```

The Cramer-Rao bound

How good could ANY unbiased estimator possibly be? That is the Cramer–Rao bound, and both layers of it are computable from things already in hand. **Per station (delay):** $\text{var}(\hat{\tau}) \geq 1/(8\pi^2\beta^2 \cdot E/N_0)$, where β is the waveform's rms bandwidth and E/N_0 the received echo energy over the noise density – bandwidth and energy are the only two currencies delay accuracy trades in (the single tone fails exactly here: $\beta \approx 0$). Monostatic ranging halves the delay error into range: $r = c\tau/2$. **Position:** each station contributes information only ALONG its look direction u_i ; the Fisher information matrix is $J = \sum_i u_i u_i^T / \sigma_{r_i}^2$ and J^{-1} lower-bounds the position covariance. Geometry enters through the u_i : directions the stations do not span are directions the network cannot measure – watch the vertical axis.

The Cramer-Rao bound — type this

```
def rms_bandwidth_hz(burst, sample_rate):
    spectrum = np.abs(np.fft.fft(burst)) ** 2
    freq = np.fft.fftfreq(burst.size, 1.0 / sample_rate)
    center = np.sum(freq * spectrum) / np.sum(spectrum)
    return math.sqrt(np.sum((freq - center) ** 2 * spectrum)
                    / np.sum(spectrum))

def range_crb_m(params: BaseStationParams, burst, echo):
    """Per-station range standard-deviation bound (meters)."""
    beta = rms_bandwidth_hz(burst, params.bandwidth_hz)
    # E/NO: echo energy over noise density, in sample units.
    energy_over_n0 = np.sum(np.abs(echo) ** 2) / params.noise_power_w
    var_tau = 1.0 / (8.0 * math.pi**2 * beta**2 * energy_over_n0)
    return (SPEED_OF_LIGHT / 2.0) * math.sqrt(var_tau)

def position_bound(stations, drone, range_sigmas):
    """Cramer-Rao covariance for the position (3x3)."""
    vectors = drone - stations
    units = vectors / np.linalg.norm(vectors, axis=1)[:, None]
    fisher = np.zeros((3, 3))
    for u, sigma in zip(units, range_sigmas):
        fisher += np.outer(u, u) / sigma**2
    return np.linalg.inv(fisher)
```

Localize the drone, and check against the bound

200 full trials: four noisy matched-filter ranges, one Gauss–Newton solve each; then measured error next to the bound, axis by axis. Measured: per-station range bound 21 cm; position **spread** $x/y/z = 0.16/0.25/0.49$ m against bounds $0.14/0.18/0.43$ m – the estimator runs within 20–40% of the theoretical limit on every axis. Two lessons in those numbers. **Geometry**: the vertical bound is $3\times$ the horizontal ones before a single trial runs – all four stations sit below the drone in nearly a plane, so their look directions barely span the vertical. **Bias is not spread**: on top of the spread sits a systematic error ($x -0.34$, $y +0.24$, $z +3.4$ m) from the ground-bounce path fusing with the direct one inside the resolution cell. The bound only governs an unbiased estimator's spread; multipath is a modeling error, it does not average away, and the weak axis amplifies it – the same lesson as the ghost target, in miniature.

Localize the drone, and check against the bound — type this (1/2)

```
def localization_study(params: BaseStationParams, trials=200,
                      out_dir="figures"):
    import matplotlib
    matplotlib.use("Agg")
    import matplotlib.pyplot as plt

    burst = make_ofdm_burst(params)
    legs = localization_legs(params)
    initial = np.array([150.0, 100.0, 120.0]) # off the plane!

    # Per-station bounds from one clean echo each.
    sigmas = []
    for gains, delays in legs:
        _, echo = measure_range(params, burst, gains, delays)
        sigmas.append(range_crb_m(params, burst, echo))
    bound = position_bound(STATIONS, DRONE, sigmas)

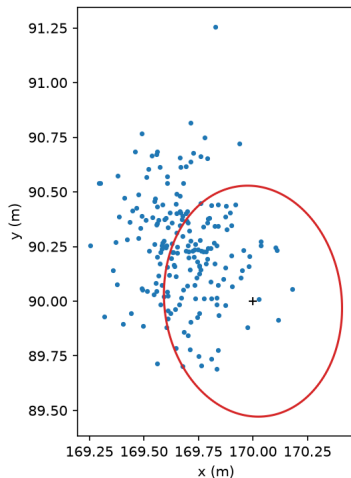
    estimates = []
    for _ in range(trials):
        ranges = np.array([
            measure_range(params, burst, gains, delays)[0]
            for gains, delays in legs])
        estimates.append(solve_position(ranges, STATIONS, initial))
    estimates = np.array(estimates)
    errors = estimates - DRONE
```


Localize the drone, and check against the bound — type this (2/2)

```
print("localization of the drone (200 trials, 4 stations):")
print(f"  per-station range bound : "
      f"{np.mean(sigmas)*100:.1f} cm")
# The bound limits the SPREAD of an unbiased estimator; the
# multipath bias is a separate, systematic effect -- report both.
for axis, name in enumerate("xyz"):
    print(f"  {name}: bias {np.mean(errors[:, axis]):+7.3f} m"
          f"  spread {np.std(errors[:, axis]):6.3f} m"
          f"  | bound {math.sqrt(bound[axis, axis]):6.3f} m")

# Scatter in the horizontal plane with the 3-sigma ellipse.
os.makedirs(out_dir, exist_ok=True)
fig, axis = plt.subplots(figsize=(6, 5))
axis.scatter(estimates[:, 0], estimates[:, 1], s=6)
values, directions = np.linalg.eigh(bound[:2, :2])
angle = np.linspace(0.0, 2.0 * np.pi, 200)
circle = np.stack([np.cos(angle), np.sin(angle)])
ellipse = (directions
           @ np.diag(3.0 * np.sqrt(values)) @ circle)
axis.plot(DRONE[0] + ellipse[0], DRONE[1] + ellipse[1], "C3")
axis.plot(DRONE[0], DRONE[1], "k+")
axis.set_xlabel("x (m)")
axis.set_ylabel("y (m)")
axis.set_aspect("equal")
fig.tight_layout()
fig.savefig(os.path.join(out_dir, "localization_scatter.png"),
            dpi=150)
return sigmas, bound, errors
```

Localize the drone, and check against the bound



200 position estimates (horizontal plane), the true position (+), and the bound's 3-sigma ellipse. The cloud's SIZE matches the ellipse – the spread is near the bound – but the cloud sits displaced from the truth: that offset is the multipath bias, and no amount of averaging shrinks it.

Run Part II

Everything measured in one run (exact output below the code). Training fields: 500/500 exact timing, 1.03 kHz frequency error, 0.107 rad phase spread. Monostatic sensing, line-of-sight: 2 paths (direct + ground bounce), echo -97.5 dBm – below the -94 dBm floor, saved by the matched filter's 32 dB – detection 100%, range error -1.2 ± 0.03 m (the small bias is the ground bounce's slightly longer path pulling the peak). Blocked: the solver finds **zero** paths – no echo exists at all, detection 0%; at these scales blockage is total, not a few dB. Blocked + reflector: the detour restores an echo (-109.2 dBm, detection 54%) but the measured range is $+37.4 \pm 1.2$ **m wrong** – a ghost target at the detour's length. One station cannot tell a detour from a distance: the concrete argument for a *distributed* array that looks from several places at once. Waveforms at 10 dB: OFDM delay error 0.0 samples rms; the smoothed tone 2.8 samples (21 m), all of it from the envelope; frequency: both at the few-kHz level set by the $40\ \mu\text{s}$ burst duration.

Run Part II — type this

```
if __name__ == "__main__":
    params = BaseStationParams()
    print(f"noise floor: "
          f"{10*math.log10(params.noise_power_w)+30:.1f} dBm")

    print("training fields (500 trials, 15 dB, 20 kHz offset):")
    test_training_fields(params)

    burst = make_ofdm_burst(params)
    for scenario in ("los", "blocked", "reflector"):
        rt, scene = monostatic_scene(params, scenario)
        _, gains, delays = monostatic_legs(rt, scene)
        rate, errors, echo_power = monostatic_range(
            params, burst, gains, delays)
        echo_dbm = (10 * math.log10(echo_power) + 30
                    if echo_power > 0 else float("-inf"))
        line = (f"monostatic {scenario:10s}: {gains.size} paths, "
                f"echo {echo_dbm:6.1f} dBm, "
                f"detection {rate*100:3.0f} %")
        if errors.size:
            line += (f", range error {np.mean(errors):+.1f} "
                    f"+- {np.std(errors):.1f} m")
        print(line)

    print("waveforms (300 trials, 10 dB, 5 kHz offset):")
    compare_waveforms(params)

    localization_study(params)

    print("rendering scenes and range profiles into figures/ ...")
    render_scenes(params)
    range_profile_figure(params)
```

Part II — measured output

The exact output of `build_sensing.py` when this guide was generated:

```
noise floor: -94.0 dBm
training fields (500 trials, 15 dB, 20 kHz offset):
  timing exact      : 500/500
  cfo error rms     : 1030.2 Hz (0.294 ppm)
  phase spread rms  : 0.1068 rad
monostatic los      : 2 paths, echo -97.5 dBm,
  detection 100 %, range error -1.2 +/- 0.0 m
monostatic blocked  : 0 paths, echo -inf dBm,
  detection 0 %
monostatic reflector : 3 paths, echo -109.2 dBm,
  detection 54 %, range error +37.4 +/- 1.2 m
waveforms (300 trials, 10 dB, 5 kHz offset):
  ofdm cfo rms      2340.9 Hz | delay rms 0.0 samples (0 m)
  tone cfo rms       7108.6 Hz | delay rms 2.8 samples (21 m)
localization of the drone (200 trials, 4 stations):
  per-station range bound : 21.2 cm
x: bias -0.318 m spread 0.163 m | bound 0.136 m
y: bias +0.245 m spread 0.244 m | bound 0.176 m
z: bias +3.441 m spread 0.507 m | bound 0.427 m
```

Where to go from here

- **Break your own builds** (you can, they are your files): feed Part I's filter only the forward capture — it locks to the channel phase, confidently wrong. Delete the π check — about half of seeds settle at zero gain reporting lock. In Part II, replace the tone's smooth ramp with a rectangular edge and watch it “gain” range accuracy it does not deserve.
- **The same code at N stations:** `ota_sync/network.py` (star), `microsync.py`, `scheduled.py`, `dfpc.py`, `hybrid_calibration/` — all built on the classes you wrote in Part I.
- **What it leads to:** `topology_selection/` — at N stations you choose which pairs exchange and how corrections apply, and those choices are not separable. And Part II's blocked-echo result is the argument for sensing with a *distributed* array in the first place.
- Regenerate this guide after any change: `python3 extract_build.py && python3 gen_slides.py && tectonic phase_sync_coding_guide.tex`